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Are Elite Women Less Corrupt?

A quantitative analysis of the relationship between women in top bureaucratic positions and corruption using perception- and scandal-based measures.

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Declaration of AI

For this thesis I have solely used AI, such as Claude AI, for code assistance in Stata and interpretation of PO outcomes with FE approach, as well as some guidance for polishing the language.

Abstract

Corruption remains a longstanding challenge in international administration and global governance. One expanding area of research concerns the relationship between gender and corruption, showing that greater female representation in government decreases corruption. While this literature tends to focus on women in politics, other institutional spheres, such as the bureaucracy, are less explored despite the fact that corruption is also widespread there. This thesis addresses this gap, examining women in top bureaucratic positions and corruption. The study employs a longitudinal cross-country design using linear regression with year and country fixed effects. Corruption is operationalized by a perceptions-based expert measure, and a scandal-based measure capturing actual involvement in corruption from Brazil and Spain. This thesis suggests that the relationship between gender and corruption is context-dependent, and the effect of women in the bureaucracy on corruption is far less straightforward compared to female politicians. Firstly, this study finds no strong support for that more women in decision-making roles within the bureaucracy reduce corruption. Secondly, this study finds no compelling evidence that their presence affects the extent to which women engage in corruption, neither in a less corrupt country, such as Spain, nor in a more corrupt country, such as Brazil. The findings challenge underlying mechanisms about the notion that women are inherently the cleaner sex, indicating that gender alone may be insufficient to explain how to mitigate corruption, and that in the bureaucracy there may be other institutional factors more at play.

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1. Introduction

In 2025, Transparency International's annual report revealed worrying trends of democratic backsliding, increasing corruption and significant declines in countries' anti-corruption efforts, (Transparency International 2026). Corruption is a global and persistent challenge, undermining legitimacy, hampering economic growth and increasing inequalities. Bribery, clientelism, money laundering, and kickbacks are some common examples of corruption (Rose-Ackerman & Palifka 2016). Scholars and policy makers have proposed a wide range of measures aimed at combatting corruption, including international agreements and soft law (Lohus & Gutterman 2021). However, enacted policies are simply not sufficient as national sovereignty enables governments to decide on the extent to which these measures should be implemented and enforced (De Vries & Sobis 2016; Lohaus & Gutterman 2021). Consequently, there is no quick fix to corruption.

In the early 2000s, a new strand of research on corruption emerged, emphasizing how corruption might be reduced by examining it in relation to gender. Preliminary studies found that higher female representation in parliament is associated with lower levels of corruption (Dollar et al., 2001; Swamy 2001). In response, researchers sought to explain this relationship. Scholars suggested that the explanations for women's effect on corruption are essentially grounded in traditional stereotypes about men and women. Women are supposed to be more honest and are held to higher standards (Dollar et al., 2001; Swamy et al., 2001; Alexander 2023), more-risk-averse (Bord & O'Connor 1997; Byrnes et al., 1999; Esaray & Chirillo 2013), excluded from corrupt networks (Barnes & Beualieu 2014; Goetz 2009), or that they simply hold different priorities and preferences (Bauhr 2024; Stensöta 2024). Furthermore, scholars note there is a significant gender gap in punishment of corrupt officials, and female politicians tend to receive significantly harsher sanctions compared to their male counterparts, leading them to avoid corruption because the penalties would be higher (Batista 2020; Dulani et al., 2021; Reyes-Housholder 2020; Woo et al., 2025).

However, the scholarship on this connection between more women in power and less corruption is divided. While many studies mentioned above show evidence of this connection, other studies do not. One strand of the literature finds that there is no gender gap in corrupt behavior in contexts entrenched by corruption, and that women do not necessarily reduce corruption (Alhassan-Alolo 2007; Alatas et al., 2009). Hence, the relationship between gender and corruption may be conditional and context-dependent rather than universal (Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017).

Most work on gender and corruption has been devoted to evaluating women in the legislature specifically. Yet, a growing body of work has examined women's role beyond the electoral-political sphere, emphasizing that corruption is prevalent among bureaucrats, too (Alhassan-Alolo 2007; Brierley & Pereira 2023; Carreras et al., 2024). Previous research stresses the importance of evaluating corruption among bureaucrats since they exhibit significant influence over core functioning's of the public sector, which are also vulnerable to corruption (Oliveros and Schuster 2018). The available evidence, although limited, has not deeply explored the connection between women in the bureaucracy and corruption (Stensöta et al., 2015). Stensöta et al. (2015) argue that women in the bureaucracy may not impact corruption levels like women in the legislature because bureaucrats lack the legislative instruments to enforce anti-corruption policies compared to politicians. However, like female politicians, female bureaucrats in senior and leadership positions remain underrepresented across the world (UNDP 2021). The gender gap at the upper ranks of the bureaucracy, which is shaped by patriarchal structures and gender biases, creates barriers to women's career advancement (Bishu et al., 2023). While these women in higher level positions do not exert political influence to enforce laws against corruption, these women have decision-making authority within the bureaucracy, affecting public service provision and organizational norms daily (Heilman 2012). Thus, women's presence in the bureaucracy, not just in the legislature, could impact corruption levels, too.

To my knowledge, this perspective of gender and corruption has not been considered in the literature before. Furthermore, there is a lack of research assessing whether and how these high-profile women in the bureaucracy engage in corruption. Therefore, this thesis aims to address those gaps with the following research questions:

- *To what extent does women's representation in top bureaucratic positions affect corruption?*
- *How does women in top bureaucratic positions influence women's involvement in corruption?*

This thesis assesses the relationship between women in top bureaucratic positions and corruption using perception-based measures and actual instances of corruption. The study is employed using quantitative analysis with a longitudinal research design. The empirical strategy relies on linear regression with year and country fixed effects. The analysis is divided into two components. Part one evaluates at a broader level the association between women in top bureaucratic positions and corruption using cross-country and cross-year data on women's presence in bureaucracies around the globe and expert survey measures of corruption. Part two uses data on corruption scandals in Brazil and Spain to examine corrupt behavior among female bureaucrats.

Overall, this thesis suggests that the effect of women in top bureaucratic positions on corruption is conditional and may only hold under certain circumstances, such as in less democratic countries. However, this evidence is not robust once controlling for both global shocks and country specific characteristics. Furthermore, analyzing the effect on women's actual involvement in corruption across two contrasting contexts, a low-corruption country like Spain and a high-corruption country like Brazil, does not produce results of a strong connection between women in the bureaucracy and corruption, as neither case separately nor together demonstrates statistically significant effects. This thesis does not find robust evidence that a greater presence of women in the bureaucracy is associated with less corruption, casting doubt on the theoretical expectation that increasing women's presence should clean up governance, at least in the bureaucratic sphere.

This thesis is structured as follows: the following section displays contemporary research on gender and corruption, followed by theory and hypotheses. The fourth section presents methodology whereas the fifth reports the analysis. The sixth section discusses the results, and the final section concludes with suggestions for future avenues.

2. Literature review

The literature review consists of four elements. The first section presents preliminary work on corruption and leads into the second part discussing gender and corruption. Drawing on existing gaps in the literature, the third section addresses corruption in the bureaucracy. The fourth and last section emphasize the role of women in top bureaucratic positions.

2.1 Corruption

Corruption has far-reaching detrimental effects on societies, eroding trust and accountability, limiting growth and economic development (Rose-Ackerman & Palifka 2016: 36), undermining competition, and increasing inequalities (Dye 2007: 7). There is a considerable amount of research on corruption, and many attempts have been made to define the phenomenon. The most standard conceptualization is the one proposed by Kaufmann (1997:114), ‘the abuse of public office for private gain’. However, corruption is understood as a multifaceted and complex concept, manifesting in multiple ways. Rose-Ackerman and Palifka (2016:9) suggest bribery, clientelism, embezzlement, extortion, kickbacks, and patronage are typical forms of corruption. Corruption operates at various levels, and a widely used distinction is between *petty* and *grand* corruption. While the former addresses small-scale transactions in public service delivery, such as bribery between street-level bureaucrats, the latter suggest collusion and complicity at the highest level of government involving political officials (Bauhr et al., 2019; Bauhr 2024).

Scholars suggest corrupt behavior may be understood under the theoretical framework of collective action problem (CAP) (Persson et al., 2013). CAP suggests that in environments where individuals are socially expected to be corrupt, individuals will continue to engage in corruption because it is collectively more difficult to do the reverse, creating an equilibrium of corrupt behavior (Persson et al., 2013: 456-457). Other studies point to structural and institutional determinants, such as the level of democracy. Previous studies show that while old and well-functioning democracies are often considered less corrupt, young democracies tend to be even more corrupt than autocracies (Bäck & Hadenius 2008:17). Young democracies often lack sufficient requisites for democracy, such as the rule of law and political competition (Keefer 2007: 804). Hence, during democratization, young democracies are particularly

vulnerable to self-reinforcing clientelism and patronage (Bauhr & Grimes 2021: 191). In response, scholars note the importance of institutional systems that enhance electoral democracy and accountability, reducing incumbents' possibilities to diffuse responsibility which could otherwise enable corruption (Charron 2011: 596)

At a global level, scholars and policy makers have proposed strategies to curb corruption. Transparency in government is an overarching principle to combat corruption (Kaufmann 1997; Dye 2007; Charron 2011; De Vries & Sobis 2016; Bauhr & Grimes 2021; Lohaus & Gutterman 2021). Transparency maintains checks and balances, ensuring independence of the judiciary, media, and civil society. Yet, these mechanisms may lose their efficiency in contexts where corruption is imposed from the top-down (De Vries & Sobis 2016: 260). Another suggestion is soft law and international agreements, such as the United Nations Convention against Corruption (Lohaus & Gutterman, 2021). While the aim is to regulate corruption by ensuring ratification and implementation by as many countries as possible, due to state sovereignty, these soft laws often fail to trigger fundamental improvements in practice (Lohaus & Gutterman 2021: 499, 504).

There is no simple panacea for corruption. Not only is corruption often hidden and illicit, but it also rarely leaves any evidence behind (Dye 2007: 8). The absence of reliable and comparable data on corruption poses a significant challenge for scholarly work trying to assess the concept. While scholars have made attempts to understand how to mitigate corruption, one potential way that has received much attention in both research and among policymakers over the last two decades is the inclusion of women in politics. The expansive research on gender and corruption explores how women's engagement in politics can deter corrupt behavior.

2.2 Gender and corruption

Across the world, women's representation in politics remains disproportionately low, especially in the Global South. The lower participation of women is influenced by institutional factors such as electoral systems, socioeconomic barriers such as poverty and limited access to healthcare and education, and cultural norms restricting women to traditional roles (Norris & Inglehart 2001:129-130.). Corruption disproportionately affects women in their daily lives and hinders their access to the political sphere (Barnes & Beaulieu 2024: 10). In the early 2000s,

researchers presented important new evidence that more women in government had a significant and negative effect on corruption (Dollar et al., 2001; Swamy et al., 2001). These findings sparked a new line of research in the scholarship of corruption, focusing on gender and the role of women. Studies on gender and corruption have grown immensely in recent years, documenting across various contexts how greater female representation in the legislature is associated with lower levels of corruption (Alexander & Bågenholm 2018; Bauhr et al., 2019; Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017; Sung 2003; Stensöta et al., 2015). However, the mechanisms underlying the relationship remain poorly understood (Alexander & Bågenholm 2018: 172). The literature has identified at least four plausible explanatory frameworks for women's lower propensity for corruption, and each factor is discussed below.

2.2.1 Conventional expectations of honesty

In the foundational studies on gender and corruption, scholars found that the significant association between higher female participation and reduced corruption held both in the legislative arena (Dollar et al., 2001) and in the private labor force (Swamy et al., 2001). One possible explanation for these findings is based on conventional expectations that women are more honest, reflecting persistent gender stereotypes. Rather than making empirical claims about inherent characteristics of women, the “fairer sex” hypothesis suggests that women may be less corrupt because they are expected to be more honest and ethical due to traditional roles, such as primary caregivers of children, and exemplifies social expectations holding women to higher standards (Dollar et al., 2001: 427; Swamy et al., 2001: 52). Therefore, Dollar et al., (2001: 424) suggest it is plausible that more women in powerful positions, such as in parliament, may have a positive impact on the behavior of their male colleagues by discouraging and limiting corrupt behavior (Dollar et al., 2001: 424). Alexander (2023: 563) suggests that if women are held to higher standards and expected to not engage in corruption since people may punish them more harshly for it, there is an increased likelihood of women becoming aware of the differential treatment, increasing women's reluctance to engage in corruption even more. However, critics of the “fairer sex” hypothesis argue that the relationship between gender and corruption cannot be explained independently of strong liberal structures. Sung (2003: 716) finds that the relationship only holds when mediated by liberal institutions that promote and enhance women's political advancement. Hence, although the fairer sex

hypothesis offers an interesting perspective, it lacks sufficient empirical evidence to be the sole explanation as to why women's presence can be associated with lower corruption levels.

2.2.2 Risk aversion

The literature has also examined whether the relationship between gender and corruption depends on women being more risk-averse than men. Drawing on broader preliminary work, there is a significant gender gap in risk tolerance, confirmed across various contexts. For example, previous research in environmental behavior shows that women are more vulnerable and worried about environmental risks (Bord & O'Connor 1997: 837), whereas studies in finance suggest that women are more cautious about their investment choices (Watson & McNaughton 2007: 52-53). However, the gender gap in risky behavior also varies over time. Byrnes et al., (1999: 377) find that, on average, men are more risk taking than women in high school, consuming alcohol and drugs, whereas in college, women become more alike men. Moreover, the literature suggests that perceived gender differences in risk-taking may be shaped by stereotypes that women would have lower self-confidence, rather than actual behavior differences (Croson & Gneezy 2009). Croson and Gneezy find that among business managers in the private sector; there is no perceived gender difference in risk-taking. While women are disproportionately underrepresented in these high-levels positions, they adapt their behavior and exhibit similar risk preferences like men (Croson and Gneezy (2009: 454).

Applying these insights to the literature on gender and corruption, scholars argue that corrupt practices which involve risk and potential reputational costs may be less attractive to women, lowering their propensity to engage in corruption (Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017). For example, Esaray and Chirillo (2013) show that greater individual risk-aversion is commonly attributed to women and is associated with a lower propensity for corrupt behavior. Yet, this finding is highly contextual since they only notice a significant effect in democracies but not in autocracies (Esaray & Chirillo 2013: 369). In democracies, corruption is culturally inimical, and democratic institutions have incentives to suppress and stigmatize corruption to reduce incentives for unethical behavior (Esaray & Chirillo 2013: 369). In accordance, Esaray and Schwindt-Bayer (2017: 662-663) argue that in democracies with effective accountability mechanisms that increase the likelihood of corruption being detected and punished, women's reluctance to corruption increase as they want to stay clean and avoid

getting caught or punished. However, more recent work by Esaray and Schwindt-Bayer (2019: 1732) revealed that in contexts with higher levels of corruption, the share of women in parliament declines simultaneously, indicating the possibility of a reversed relationship. Nonetheless, Alexander (2023: 562) suggest that women may be more risk-averse in democracies whereas in authoritarian regimes women may engage in corruption because refusing to might entail greater risks.

The literature contend that risk-aversion may only partially explain gender differences in corrupt behavior since studies suggests that elite women with powerful positions in male-dominated sectors may be less risk-averse (Bauhr et al., 2019: 1047). Previous work on female mayors in France exemplify this (Bauhr & Charron 2021). While newly elected female mayors had a significant negative effect on corruption, reelected female mayors did not, indicating that they may be more prone to corruption because of marginalization and exclusion. Once they attain positions of power, they may adapt and navigate corrupt networks to ensure their tenure (Bauhr & Charron 2021: 1313). In addition, previous research also proposes that women's greater risk-aversion may not be professed in environments where corruption is endemic, suggesting that there is no gender gap in neither engagement nor tolerance for corruption (Alatas et al., 2009; Brierley & Pereira 2023; DiRizenzo 2019). For example, Alatas et al., (2009) field an experiment in Australia, India, Indonesia and Singapore that shows that only women in Australia expressed strong condemnation of corruption, suggesting that tolerance for corruption may be strongly shaped by cultural and contextual factors rather than by gender alone. Similarly, Brierley and Pereira (2023:4) could not find any gender distinction in paying a bribe to bureaucrats among citizens in Ghana. Moreover, DiRizenzo (2019) suggests that corrupt behavior may not necessarily be strongly associated with personal traits but rather culture. She finds that more individualistic societies tend to exhibit less corruption as such societies are more likely to uphold social norms that condemn corrupt practices, and women are more likely to adhere to them, plausibly because they are more risk-averse. Contrarily, collectivistic cultures exhibit more corruption. DiRizenzo suggests that collectivistic cultures have more social expectations that encourage bending the law to benefit in-group members. Hence, the potential anti-corruption effect of women's representation is weakened since social norms may permit violations if they benefit the group to which a female public official belongs, indicating that these women may not be very risk-averse (DiRizenzo 2019: 328).

2.2.3 Outsider status

There has also been considerable interest in whether women are less corrupt because they are more often excluded from corrupt networks. Feminist scholarship suggests that gendered traits, such as women's perceived morale and honesty, have historically been used to justify exclusion of women from the political sphere (Goetz 2009:90). This omission is rooted in entrenched gender systems that subordinate women to men and reinforce male-dominated networks, which can foster corrupt practices (Ottervik & Zheng-Su 2023: 623). Consequently, Ottervik and Zheng-Su (2023: 644) find that patriarchy is a significant predictor for both greater acceptance of corruption and disempowerment of women. Furthermore, since corrupt systems are predominantly male and men act as gatekeepers to control access to positions and resources, as a result, in highly corrupt environments these corrupt circles increase systematic exclusion of women (Stockemer & Sundström 2019: 87). Hence, the literature suggest that women's lower propensity for corruption may be explained by the fact that women are often positioned as outsiders to informal structures through which corruption operates.

The available evidence supports this claim. Previous work on female councillors in Spain suggests that because of women's outsider status, they are less likely to engage in corruption (Wieseholmer & Verge 2020: 546). Likewise, Pessa et al., (2025:8) note a significant gender gap in corrupt networks in Brazil and Spain, with women constituting a minority of actors within corrupt networks, approximately ten and twenty percent, respectively. In Russia, women's exclusion from corrupt networks also increases citizens reluctance to initiate corrupt practices with female bureaucrats (Zaloznaya et al., 2024: 759-760). Furthermore, prior work also suggests that women's exclusion may be driven by their greater risk-aversion (Barnes et al., 2018), and that because women are often described as a marginalized group, they may not be entitled to the institutional power or opportunities necessary to practice corruption (Alexander 2023: 562). For instance, Barnes et al., (2018:155) find that both risk-aversion and outsider status are credible predictors for why female police officers produce greater trust in the police, and that hiring more women reduces corruption.

However, as mentioned earlier, elite women may underline an important exception, both in terms of risk-aversion and outsider status, and may be more prone to corruption once they have access to corrupt networks (Bauhr & Charron 2021). Wieseholmer and Verge (2020: 545-546)

argue that although female councillors are less willing to engage in bribery, with longer tenure, they are also deemed to become more corruptible if maintaining close ties to entrepreneurs and businesses in the private sector. Another reasoning suggests that the exclusion of women from corrupt networks may encourage women to mobilize against corruption or possibly motivate them to create their own (Bauhr et al., 2019: 1048). Consequently, the empirical evidence implies that women's lower propensity for corruption cannot be explained solely by a fairer sex hypothesis but also must consider women's risk-aversion and their exclusion from corrupt networks.

2.2.4 Interests mechanisms

The literature on gender and corruption also suggests that women may be less inclined to engage in corruption because their policy priorities are shaped by systematic gendered differences. In public administration, women are overrepresented in traditionally feminized policy areas, such as health, education, and labor (UNDP 2021: 67). Contrary to men, women in the electorate often prioritize policies aimed at improving public service delivery and social welfare (Bauhr et al., 2019: 1047). Consequently, female officials may refrain from engaging in corruption since such behavior would undermine these policy priorities that they seek to preserve and protect (Bauhr 2024; Stensöta 2024). Furthermore, Barnes and Beaulieu (2024:5) find that women are more likely to work in policy areas with fewer opportunities for corrupt behavior, whereas men are overrepresented in sectors, such as finance, which may present greater opportunities for corruption, hence women's lower involvement in corruption could simply be due to gendered occupational sorting. Also, Evidence from Latin America show that female legislators are more concerned with corruption than men and overrepresented in advocating for anti-corruption policies (Alexander & Bågenholm 2018: 183). In response, female candidates may enjoy an electoral advantage when campaigning against corruption. Barnes and Beaulieu (2014: 366) show that voters gain more trust in government when more female officials are elected, and perceptions of electoral malfeasance decline simultaneously. Similarly, Guajardo and Schwindt-Bayer (2025: 1532) find that, following corruption episodes, female candidates in municipal elections in Mexico are significantly more likely to win than male candidates.

2.2.5 Consequences of corruption

Recent work has also examined the consequences of corrupt women in the legislative, indicating a significant gender gap. Contrary to men, corrupt women face significantly worse consequences, such as greater reputational costs, low levels of trust or plummeted approval ratings, and higher probability of being dismissed (Dulani et al., 2021; Reyes-Housholder 2020; Woo et al., 2025). For instance, Carlini et al., (2020: 1371) show that female presidents enter with less support than male presidents, and that following corruption scandals, public approval of female presidents drops significantly whereas for male presidents there is little to no comparable effect. Similarly, Reyes-Housholder's (2020: 545-546) find that both President Dilma Rousseff in Brazil and President Michelle Bachelet in Chile received greater backlash and distrust, and even impeachment of Rousseff, due to corruption related controversies compared to their (corrupt) male predecessors and successors. Likewise, studies in Malawi (Dulani et al., 2021), in Mexico (Batista 2020), and in South Korea (Woo et al., 2025) confirm that only female incumbents face significant decline in reelection prospects after scandals. Consequently, the literature suggests that these findings reflect persistent gender norms about women. When women engage in corruption, they are perceived to deviate more strongly from expected behavior, and, as a result, women experience greater disapproval and punishment compared to men (Reyes-Housholder 2020; Batista 2020: 1566). Even beyond the political sphere, prior work in the private sector notes a significant gender punishment gap in misconduct since female managers are one-fifth more likely to be dismissed compared to their male colleagues (Egan et al., 2022: 1186). Nevertheless, literature is not unanimous. Eggers et al. (2015) examination on electoral punishments of corrupt politicians finds no significant gender gap. However, they note that female voters tend to punish female politicians more harshly, plausibly because they hold other women to higher standards (Eggers et al., 2018: 325). Prior work suggests that voters may be more forgiving of corrupt politicians when the corrupt practices are perceived to serve the public good, such as job creation (Le Foulon & Reyes-Housholder 2021:3). Le Foulon and Reyes-Housholder's (2021:9) find that in Uruguay male politicians face harsher punishment, yet in Argentina and Chile, voters still penalize women more severely, consistent with broader findings in the literature.

2.2.6 Identifying a gap

The literature on gender and corruption proposes four key explanations for women's lower propensity to engage in corruption: the "fairer sex" hypothesis, greater risk aversion, exclusion from corrupt networks, and different political interests. Most work on gender and corruption has focused on elected officeholders, suggesting that greater share of female representation is associated with less corruption, at national level in parliament (Alexander & Bågenholm 2018; Armstrong et al., 2022; Dollar et al., 2001; Swamy et al., 2001; Stensöta et al., 2015), at executive level in cabinet positions (Stockemer & Sundström 2019), and at local level (Bauhr et al., 2019: 1044; Bauhr & Charron 2021: 1309). The available evidence also suggests that the relationship between women and corruption may be reversed (Esaray & Schwindt-Bayer 2019) and that it relies on contextual factors and institutional factors since the gender gap disappears in environments where corruption is normalized (Alatas et al., 2009; DiRizenzo 2019; Esaray & Chirillo 2013; Brierley & Pereira 2023). Moreover, most of these studies are only able to show that increased female participation is associated with lower levels of corruption without directly testing the underlying mechanisms. Consequently, the evidence points to that the relationship between gender and corruption relies on broader structural factors, such as the level of democracy and the presence of liberal institutions, which both promotes women's advancement and constrains corruption. Nevertheless, the scholarship on gender and corruption has predominantly evaluated the behavior of elected politicians, yet we know that bureaucrats engage in corruption too. Thus, the next section addresses this further.

2.3 Corruption in the bureaucracy

The literature on gender and corruption in the bureaucratic realm is inconclusive. The available evidence, albeit quite limited, indicates that women's involvement in corruption in the bureaucracy hinges on contextual factors (Alhassan-Alolo 2007; Brierley & Pereira 2023; Carreras et al., 2024). Previous findings in Ghana show that once women are exposed to corrupt networks and collectivistic cultures, female bureaucrats are often equally corrupt as men (Alhassan-Alolo 2007: 236-237). Similarly, Alatas et al., (2009) highlight the significance of context and culture as their cross-country study on bribery found no consistent gender difference in tolerance for corruption in the Asian countries examined. Likewise, recent work on bureaucratic corruption in Ghana and Paraguay suggest that once there is no stigma attached to corruption, citizens does not expect any gender distinctions in corrupt behavior among

bureaucrats (Brierley & Pereira 2023; Carreras et al., 2024). Consequently, the competing evidence from the legislature versus the bureaucracy complicates the ability to draw conclusions on whether women are less corrupt than men in all governance institutions or only in select institutions. Do female and male bureaucrats also engage in corruption differently?

Corruption at the administrative level typically signifies petty corruption, such as small-scale, informal and illicit transactions between bureaucrats and their clients, particularly prevalent in developing and non-democratic countries (Zaloznaya et al., 2024: 740). The available evidence suggests that wages are a significant factor behind civil engagement in corruption. Graeff et al. (2025: 14-15) find that increasing public officials' salaries has a significant effect on reducing their corruptibility. Furthermore, public servants are more likely to engage in corruption if there is a lack of job turnover. Turnover disrupts mutual trust among co-conspirators by rotating and forcing change in exploited corrupt networks of corrupt offices (De Vries & Sobis 2016). However, even with turnover, relatively honest people may be persuaded to become corrupt due to social and organizational pressure (De Vries and Sobis 2016: 262-263). The existing evidence also indicates that the more public servants engage with private businesses, the more egoistic, unethical and susceptible to corruption they become (Graeff et al., 2025:5). For example, in Brazil, the risk of being dismissed due to corruption is far greater the more public officials are involved in private businesses (Graeff et al., 2025: 15). Lastly, politicized bureaucracies significantly heighten the risks of corruption. When bureaucrats' careers heavily depend on approval of political superiors, this dependency increases the possibility that they engage in corruption to maintain their positions (Meyer-Sahling et al., 2018: 283). Evidently, Oliveros and Schuster (2018) find that politically and personally appointed public servants are more willing to engage in corruption. While previous research on bureaucratic corruption has identified mechanisms explaining why bureaucrats and public servants may engage in corruption, these explanations are general and do not focus particularly on gender. Little attention has been given to the experiences of female bureaucrats and is further discussed below.

2.4 The *elite women* of the bureaucracy

A lot of work on gender and corruption has theoretically and empirically focused on elected officials despite evidence that bureaucrats engage in corruption, too. While the literature offers a high level of support for the finding that female politician's lower corruption, these patterns are not examined as extensively within the bureaucracy. Thus, we should care about and seek to understand what may deter female bureaucrats from engaging in corruption. Furthermore, since theories about elected women and less corruption are theoretically based in arguments about gender differences which exist across society broadly, we should expect there to be a gender gap in corrupt behavior between female and male bureaucrats, just like with politicians. Previous work argues that although bureaucrats are not making policy, they have a significant impact on governance, public service provision, but also the extent of corruption (Oliveros & Schuster 2018: 760). Moreover, Alexander (2023: 572) suggests that research on how bureaucratic environments compared to more personalistic environments, such as those of elected officials, may suppress gender differences is badly needed, yet remains understudied.

Earlier work on women in the bureaucracy tends to examine the role of women's careers rather than their susceptibility to corruption. Like women in politics, women in the bureaucracy, and particularly women in higher-level positions, are underrepresented globally (UNDP 2021: 29). Prior work suggests this disproportionality of women and male-domination at the upper ranks could depend on prescribed attributes and gender bias, creating hurdles for women's advancement (Bishu et al., 2023; Heilman 2012). For instance, a study on female senior managers in municipal governments show that women experience significant gender discrimination in the forms of greater scrutiny and constant need of legitimacy from male subordinates and superiors (Bishu et al., 2023). Furthermore, previous studies suggest the scarcity of women in higher positions within the bureaucracy increases the perception of them as tokens (Groenevald et al., 2020: 444). Nevertheless, some work, albeit limited, has examined the association between women in the bureaucracy and corruption. Initially, Swamy et al., (2001:40) suggested that increased female representation in parliament should generate positive spillover effects such as greater female participation in the bureaucracy and lower levels of corruption overall. Yet, Stensöta et al., (2015) challenges this expectation since they find no significant association between higher participation of women in the bureaucracy and corruption. The authors argue this may reflect that bureaucrats lack legislative authority and

have limited capacity to pursue anti-corruption reforms compared to elected officials (Stensöta et al., 2015: 494).

The limited attention given to women in the bureaucracy and corruption is both surprising and problematic. Insufficient attention has been given to how women in top positions within the public sector—such as those occupying senior, managerial, and other key leadership positions—respond to corruption. While these women are not tied to representative political power since they are neither elected by voters, like politicians, nor politically appointed, like cabinet ministers, these women still have significant discretion over decision-making and resource allocation within the public sector (Groenevald et al., 2020: 443). Hence, understanding their attitudes and incentives toward corruption is central to the study of gender and corruption. A growing body of work suggests that increased diversity within public institutions can improve institutional outcomes. For example, Hong (2021: 1334-1335) finds that greater ethnic minority representation in the London police force improved bureaucratic performance and public service delivery. Similarly, Barnes et al. (2018:155) show that higher share of female officials in the police force reduce suspicions of corruption, partly since women are perceived as more risk-averse and less embedded in corrupt networks. While neither study explicitly examines the role of women in senior positions within the bureaucracy, they highlight the benefits of diversity in government outcomes, such as reduced corruption. Consequently, the relationship between women in senior or top positions, the *elite* women of the bureaucracy, and corruption has not yet been adequately explored, pointing to a particularly significant gap in the gender and corruption literature.

3. Theory

In this section, I build on existing research on gender and corruption to construct a theoretical framework to assess the relationship between women in top bureaucratic positions and corruption. Based on the theories developed to explain how and why female politicians may reduce corruption, I suggest these theories are applicable to women top bureaucratic roles, referring to women in managerial, senior and top leadership positions within the bureaucracy. I argue that these women have significant power and oversight, and that these women are closer,

both professionally and policy-wise to female politicians compared to street-level bureaucrats. Contrary to street-level bureaucrats, women in top bureaucratic positions can make decisions on different aspects of the public sector, manage policies, and exercise oversight of the organization which are all influential responsibilities more akin to the power of politicians than the responsibilities of street-level bureaucrats. In the following section, I argue why contemporary theories of women's lower susceptibility to corruption transfers to these women. Secondly, I present two conceptualizations to test my theory.

3.1 Why existing theories applies to women in top bureaucratic positions

Firstly, the fairer sex hypotheses do not necessarily pertain to female elected officials exclusively. These are gendered attributes pertaining to women overall. The literature on gender and corruption (e.g. Goetz 2009; Esaray & Chirillo 2013; Barnes & Beaulieu 2014), as well as previous studies (e.g. Bishu et al., 2023) on women in the bureaucracy, both highlight the role of gender stereotypes. In the former, women are often perceived as less corrupt due to expectations of greater morale and honesty, whereas in the latter such stereotypes are identified as barriers to women's career advancement (Heilman 2012). I argue these perspectives are not mutually exclusive; rather, similar expectations of honesty are likely to apply to women across institutional contexts, regardless of whether they are politicians, bureaucrats or public servants. I contend that even though female politicians are elected by voters and face heightened expectations to conform to gendered norms because they operate under the threat of electoral accountability and voters' influence, there is little reason to assume that female bureaucrats would be held to very different moral standards. Either job comes with greater gender biases, either at the polls or at the workplace. I argue that there is a possibility that female bureaucrats would be less inclined to engage in corruption, not necessarily due to inherent traits, but because violating these expectations generates greater professional costs, such as losing their jobs, regardless of the absence of the electoral mechanism that impacts politicians.

Secondly, previous research suggests that women in the electoral sphere are less prone to corruption because they may be more risk-averse than men (Eсарay & Chirillo 2013; Esaray & Schwindt-Bayer 2017). However, some studies note that this explanation may not be held entirely (Bauhr & Charron 2021). Over time, elite women may become more susceptible to

corruption as they gain greater access to powerful networks where corrupt practices are normalized and reinforced by institutional pressure (Bauhr et al., 2019; Bauhr & Charron 2021). Under such circumstances, engaging in corruption may result in women securing longer tenure. At the same time, prior work suggest that women are more risk-averse in developed democracies where corruption is detected and more likely to be punished (Esaray & Schwindt-Bayer 2017). Even outside the political realm, there is evidence from the private sector pointing to significant harsher punishment and bias against women caught in misconduct (Egan et al., 2022), which adds on to the idea of why women may be more risk-averse than men. Informed by these findings, I argue that women in top bureaucratic positions may, similarly to female politicians, exhibit greater risk-aversion due to their underrepresented and vulnerable status. Since women continue to face barriers to accessing high-level positions because of gender bias (Heilman 2012) they may feel a stronger need to establish legitimacy and avoids behavior that could potentially threaten their profession. Consequently, these women may be more reluctant to engage in corruption.

Thirdly, the literature on gender and corruption suggest that women may engage less in corruption because women are often excluded from informal and male-dominated corrupt networks (Barnes & Beaulieu 2014,2018; Goetz 2009; Stockemer & Sundström 2019). Thus, it is the inaccessibility to corruption that reduces women's likelihood of engaging in corrupt rather than gendered behaviors. Like women in politics, prior research on women in senior bureaucratic positions suggest that these women are underrepresented within elite institutions (Bishu et al., 2023; Heilman 2012), implying that they are already exhibiting an outsider status. The evidence indicates that since corrupt networks are predominantly male and women constitute a minority within those, at least in the political branch (Pessa et al., 2025; Zaliznaya et al.,2025; Wieselholmer & Verge 2020) there are reasons to assume that similar dynamics exists within the bureaucracy. Prior work argues that senior bureaucratic positions are mostly occupied by men (Bishu et al., 2023; Heilman 2012), indicating that they may also be more likely to be involved in corrupt circles within the bureaucratic sphere compared to their female counterpart. Consequently, even when the presence of women at the upper ranks of the bureaucracy increases, they may still have limited access to entrenched corrupt networks, and this exclusion, like female politicians, reduce their opportunities to engage in corruption.

However, like Bauhr et al., (2019) suggest, attaining higher positions in corrupt environments may also create pressures for these women, like female politicians, to engage in corruption due to institutional norms, making them more risk-tolerant and more prone to corruption. Furthermore, previous research shows that women's risk-aversion and exclusion is not necessarily pronounced in all contexts (Alhassan-Alolo 2007; Alatas et al., 2009; Brierley & Pereira 2023; Carreras et al., 2024). Essentially, this suggests that women in top bureaucratic positions may become equally susceptible to corruption as female politicians in environments where corruption is normalized and not strongly stigmatized. Nevertheless, I suggest that the theoretical frameworks of risk-aversion and outsider status in many aspects extends to women in top bureaucratic positions, and are plausible mechanisms increasing their lower engagement for corruption within the bureaucracy.

Lastly, in response to that previous research have not been able to find a significant effect between women in the bureaucracy and corruption, scholars argue this could depend on that compared to female politicians, female bureaucrats lack political authority and do not legislate policies that directly affect corruption (Stensöta et al., 2015). Nonetheless, scholars argue that bureaucrats exert significant influence over the functioning of the public sector and public service delivery, which are certainly not immune to corruption (Oliveros & Schuster 2018). Therefore, I argue that women in top bureaucratic positions may be particularly vulnerable to corruption due to their strategically important role, which can be exploited by others. Given their minority status, these women may face greater pressure to conform to corrupt norms. However, previous research suggest that women are more likely to work in sectors less exposed to corruption, whereas men more often occupy positions in more corruption prone areas (Bauhr 2024; Stensöta 2024). As a result, women in top bureaucratic positions may remain less exposed to corruption if they remain in the more feminized institutions than if they operate in more male-dominated sectors.

Hypotheses About General Corruption

Considering previous findings that higher female participation in parliament is associated with lower levels of corruption (e.g Dollar et al., 2001; Swamy et al., 2001), this study explores the relationship between gender and corruption in a new bureaucratic context. This study argues that the mechanisms commonly used to explain corrupt behavior among female politicians

should transfer to women in top bureaucratic positions. The first part of the analysis will assess how the share of women in top bureaucratic positions (x) affects corruption (y), and the first hypothesis follows:

H¹: A greater share of women in top bureaucratic positions is associated with lower levels of corruption.

The literature on gender and corruption suggests that the negative association between more women in government and corruption hinges on contextual and institutional factors (DiRienzo 2019; Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017). Scholars have found that the association only holds in developed democracies and more individualistic cultures where corruption is condemned (Eсарay & Chirillo 2013; Esaray & Schwindt-Bayer 2017), whereas in contexts where corruption is the norm the gender gap declines or disappears entirely (Alhassan-Alolo 2007; Alatas et al., 2009; Brierley & Pereira 2023). Prior work suggests the level of democracy is a significant predictor of corruption since well-functioning democracies tend to exhibit low corruption (Bäck & Hadenius 2008; Keefer 2007). Based on their findings that greater female participation reduce corruption, Swamy et al., (2001) suggested this could eventually result in extended women's representation in public administration overall. In response, Sung (2003) find that only democracies with strong liberal institutions facilitate an effect of increased female participation in parliament which in turn reduce corruption. In line with previous findings, I contend that democracy should also condition the relationship between women in the bureaucracy and corruption. Consequently, in more democratic countries women should not only be more likely to enter the political arena but also the bureaucracy, and once present, they should also exhibit influence on corruption, like female politicians. Thus, the second hypothesis follows:

H²: A greater share of women in top bureaucratic positions will have a stronger negative effect on corruption in more democratic countries compared to less democratic countries.

Hypotheses About Involvement in Corruption

Beyond assessing the general relations between women in top bureaucratic positions and aggregate levels of corruption, it is also interesting to look closer at actual behavior. Hence,

the second part of the analysis delves into how women's presence in top bureaucratic positions (x) affects women's involvement in corruption (y). In this part, corrupt behavior will be measured in terms of corruption scandals and female involvement which is further discussed in the methodology section. Overall, I suggest that as women's representation in the public sector increases, their access to higher public sector positions should also expand over time. This development provides an opportunity to examine whether women's involvement in corruption changes accordingly. If a greater share of women appears in corruption as their representation in the public sector grows, this could support the argument that women have historically appeared less corrupt primarily because they have had limited access to positions where corruption occurs. Thus, once women gain greater access to these top public sector roles and become more integrated into administrative networks, they may also become more exposed to opportunities for corruption. If corruption increasingly involves more women over time, this will challenge the notion that women are inherently less corrupt than men due to gendered qualities like greater honesty or greater risk aversion. Alternatively, it is possible that as women gain greater access to top positions, corrupt behavior remains predominantly associated with men. If this pattern persists despite increased female representation, it suggests that women's lower involvement in corruption cannot be explained solely by outsider status or limited access to corrupt networks. Instead, it may indicate that other factors such as differences in incentives, norms, or institutional dynamics contribute to women's continued underrepresentation in corruption, which is beyond the scope of this thesis to assess.

In line with my first hypotheses which assess a general level of corruption in a given country, I expect that a greater share of women in top bureaucratic positions is associated with less corruption scandals. To test these expectations, I propose the following hypothesis:

H3: A greater share of women in top bureaucratic positions is associated with less corruption scandals.

H4: A greater share of women in top bureaucratic positions is associated with less female involvement in corruption.

Given the importance of contextual factors shaping the relationship between gender and corruption, I consider context equally central for the second analysis. I suggest that since democratic systems promote female participation, there is a greater likelihood that democracies

have a higher representation of women in top positions in the public sector. Prior work (Esaray & Chirillo 2013) suggests that democracies with lower levels of corruption has a deterrent effect on corrupt behavior, particularly among women. Democracies exhibit greater transparency, accountability, and legal enforcement against corruption, and therefore suggest that women's risk-aversion may be more pronounced (Esaray and Schwindt-Bayer 2017). Hence, I suppose that under such circumstances, in more democratic countries with low corruption, the presence of more women in top bureaucratic positions should also lead to less women inclined to engage in corruption.

Conversely, in high-corruption contexts, I argue that the recruitment of women to these top positions in the bureaucracy is plausibly less prevalent as institutions are ruled by inherently male-dominated corrupt networks. Those women who do attain such positions may engage in corruption to maintain their position. Furthermore, in conjunction with previous research (e.g Alatas et al., 2009; DiRenzo 2019; Brierley & Pereira 2023; Carreras et al., 2024) suggesting that there is no gender gap in corrupt behavior in contexts where corruption is normalized, the deterrent effect of the reputational risk is weakened, and under such conditions, this may increase the likelihood that women, especially those in powerful positions, engage in corrupt practices and the gender gap in corrupt behavior dissolves because it involves more women. Drawing on these insights, the last hypothesis follows:

H5: The association between greater share of women in top bureaucratic positions and less female involvement in corruption is stronger in low-corruption contexts than high-corruption contexts.

4. Methodology

Empirical work on gender and corruption often entails challenges. Not only is corruption an illegitimate act, but it is also a sensitive topic, and public officials have incentives to hide their corrupt behaviour. Thus, researchers must consider alternative ways to investigate the phenomenon since data on the true amount of corruption in a country does not exist (Piazza & Diaz 2024). In the preliminary work on gender and corruption, proxies for corruption, such as the international country risk guide, was used by Dollar et al., (2001) to confirm the negative link between women in parliament and corruption. While proxies for corruption, institutional

or contextual factors, such as institutional strength or trade openness, are useful in contexts where information on corruption is weak, the validity is weakened since these proxies do not explicitly capture corruption and may also reflect biases among those overseeing political institutions (Piazza & Diaz 2024: 154). Hence, most studies on gender and corruption have used perception-based measures to assess corruption, such as the Corruption Perceptions Index from Transparency International (e.g Alexander & Bågenholm 2018; Alhassan-Alolo 2007; Barnes et al., 2018; Bauhr & Charron 2021; Bauhr et al., 2019; Carlini et al., 2020; Dulani et al., 2021; Eggers et al., 2017; Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017; Guajardo & Schwindt-Bayer 2025; Jha & Sarangi 2018; Le Foulon & Reyes-Housholder 2021; Reyes-Housholder 2020; Stensöta et al., 2015; Woo et al., 2025; Wieselholmer & Verge 2020). Perceptions-based studies, especially those using expert assessments, provide more explicit and valid indicators of corruption, and are beneficial as they are generally comprehensive, representative, applicable across multiple contexts (Piazza & Diaz 2024: 148), and cost-effective (Costa 2024:140). Yet, like proxies, perceptions are not measuring direct observable outcomes of corruption and may also reflect potential ideological gendered biases or biases in terms of that self-reported attitudes do not reflect actual behavior (Piazza & Diaz 2024: 156).

4.1 Design and modeling strategy

The research design for this study relies on quantitative analysis to assess the relationship between women in top bureaucratic positions and corruption. In quantitative research, the modelling strategy often builds upon a distinction between a treatment and control group (Angrist & Pischke 2014). However, the effect of more women in the bureaucracy on corruption cannot solely be explained by an exogenous shock. Since this study explores no single treatment effect, causality is evidently challenging, and there are reasons to assume that other predictors besides women in the bureaucracy affect corruption too, such as institutional and contextual factors (Sung 2003; DiRizenzo 2019; Esaray & Chirillo 2013). Also, prior work shows that the relationship between women and corruption may in fact run in the opposite direction, suggesting reversed causality (Eсарay & Schwindt-Bayer 2019), further questioning the ability to draw inferences. Isolating the effect solely of women may not reveal causality but rather correlation. Given these challenges to causal inference, this thesis tests for significant associations between my independent and dependent variable rather than causal relationships

¹. This study utilizes observational data and employs a longitudinal cross-country design. The modelling strategy for the analysis draws upon that corruption can be measured in multiple ways: more indirectly via expert survey perceptions and more directly via data on corruption scandals.

Part one of my analysis focuses on indirect ways to operationalize corruption. Like previous research, part one use comparable perception-based measures of corruption, which is the most common, effective and applicable assessment of corruption across countries. This part of the analysis uses panel data with experts' assessments and cross-national survey data to examine the relationship between women in top bureaucratic positions and corruption between 2000 to 2022.

Part two of my analysis investigates the relationship using corruption scandals as a more direct measure of revealed corruption. Piazza and Diaz (2024: 151) argue that while audits are the most proximate measure of corruption, scandal data are particularly useful and valid indicators for capturing real instances of corruption. Yet, building datasets on scandals places significant burden on the researcher to comprehend and interpret multiple sources and their reliability. Scandal data is also often limited in its ability to generalize findings across countries since available datasets on scandals are scarce. Moreover, such reports may be influenced by disparities in power networks and media biases, which can further influence corruption propensities between men and women (Piazza & Diaz 2024: 153). Overall, there is a lack of good cross-country, cross-year, and systematic data on corruption scandals, but scandal data can still provide an informative measure of revealed corruption.

The existing research using scandal data, such as Pessa et al. (2025), suggests that only scandals expose corrupt behavior and are only observable once they are publicly disclosed. While corruption tends to be hidden and illicit, scandals are one expression of corruption that is by nature very public. The dilemma is that we do not have abundant data on this, neither across countries nor time, whereas for corruption perceptions we do. Contrary to the perceptions-based assessment in part one, the scandal data allow for examination of visibility and detection, which are factors that are central to theories of risk-aversion and outsider status. Thus, examining scandal data, this study allows for a direct test of whether the commonly proposed mechanisms

¹ A potential outcomes framework is available in appendix, see A13.

in the gender and corruption literature are reflected in actual behavior among women in top bureaucratic positions. Yet, this scandal data cannot empirically confirm whether these women are more honest since it only detects and report corruption and those included, perhaps underestimating the true extent of corrupt behavior, just like perception-based measures. Nevertheless, scandal data provides an alternative operationalization of corruption that can capture more fine-grained data about who is involved in such scandals than expert surveys can capture.

Part two is a comparative analysis of corruption scandals in Brazil and Spain between 2002 to 2014. The selection of these countries is based on the available data retrieved from an existing study (Riberio et al., 2018). These data, collected through local newspapers and media outlets (detailed description under *4.3 Data*) were provided to me by the authors of this study upon request Ribeiro et al., (2018). Brazil and Spain also provide interesting variation to test H5. Brazil is a federal democracy, yet the country suffers severely from criminality and corruption with an inefficient judiciary, undermined independent media and civil rights (Freedom House 2026a). The CPI, which is an expert assessment of country level corruption, ranging from zero, highest perceived corruption, to a hundred, lowest perceived corruption, ranks Brazil 107th place out of 182 with a score of 35 (Transparency International 2026a). Gaps in federal transfers, collusion and cartels are some examples of corrupt practices undermining Brazil democracy resilience, and in 2014 the biggest corruption scandal to date was revealed of large-scale bribery and kickbacks between state enterprises, politicians, bureaucrats and private businesses (Dominguez 2025:6). Hence, Brazil is considered a high-corruption context. Spain is also considered a free and democratic country by Freedom House, with an independent judiciary, high rule of law and respect for civil liberties (Freedom House 2026b). In comparison to Brazil, Spain has a CPI of 55, ranking it 49th place out of 182 countries (Transparency International 2026b) and most corruption events concern money laundering felonies (Carabantes et al., 2018: 39) Thus, for this study, Spain constitutes a low-corruption context.

4.2 Method

The analysis is conducted with linear regression with two-way fixed effects (TWFE). The fundamental advantage with this model is that it examines changes within-countries over time and allows for measurement of associations between variables (Bruderl & Volker 2013:3).

While year FE controls for time-variant factors, such as global shocks and trends, and country FE, which absorbs all time-invariant country characteristics, such as history and culture (Stathelp n.d-a). Contrary to the Difference-in-Differences (DiD) model which requires a distinct treatment and control group, building on a parallel trend's assumption (Angrist & Pischke 2014: 203), linear regressions with fixed effects (FE) and two-way fixed effects (TWFE) can be run without a clear experimental setting. Another key strength with this model is that it builds on panel data, which consist of repeated observations over time and produce multiple observations for each unit, increasing the analysis's statistical leverage (Fortin-Rittberger 2013: 6) and reducing collinearity among explanatory variables (Hsiao 2003:3). Moreover, heterogeneity, unobserved variables that remain constant over time and that are not explained by the independent variable, is further reduced with FE models which assume that each unit has its own intercept, and that the covariation is fixed in an intercept term rather than varying as a random variable (Fortin-Rittberger 2013:15). While exhibiting several benefits, FE models cannot examine the underlying factors causing the change within-countries (Stathelp n.d-a). Nevertheless, I contend this empirical strategy is the most suitable approach for testing the hypotheses.

The baseline estimation for panel regression with FE (H1, H3-H5) follows: $Y_{it} = \beta X_{it} + \alpha_i + \gamma_t + \epsilon_{it}$. Y_{it} equals corruption, for country i at time t . X_{it} equals women in top bureaucratic positions, α_i equals the country fixed effects and γ_t , year fixed effects, and ϵ_{it} is the error term.

The baseline estimation (H1, H3-H5) includes controls C_{it} follows: $Y_{it} = \beta_1 X_{it} + \gamma C_{it} + \alpha_i + \gamma_t + \epsilon_{it}$.

The estimation for panel regression with FE and heterogeneous effects (H2), includes the moderator (democracy) Z_{it} with B_3 capturing the interaction term:

$$Y_{it} = \beta_1 X_{it} + \beta_2 (X_{it} \times Z_{it}) + \beta_3 Z_{it} + \alpha_i + \gamma_t + \epsilon_{it}$$

Lastly, the full estimate includes interaction and controls (H2):

$$Y_{it} = \beta_1 X_{it} + \beta_2 Z_{it} + \beta_3 (X_{it} \times Z_{it}) + \gamma C_{it} + \alpha_i + \gamma_t + \epsilon_{it}.$$

4.3 Data

The analysis uses data from the Quality of Government Institute (QoG), the World Bank's Worldwide Bureaucracy Indicators (WWBI), World Development Indicators (WDI), and original collected data on corruption scandals in Brazil and Spain, supplied by Professor Riberio (Riberio et al., 2018). Part one consists of the QoG Standard time series (TS) dataset Version2016 and WWBI. The QoG Standard TS contains data between 1946 to 2025, and the unit of analysis is country-year. This data is particularly useful since QoG specializes in comparative research on quality of government and corruption (Teorell et al., 2026). The WWBI contains cross-national data on public sector employment, derived from administrative and nationally represented household surveys, conducted in waves (six versions) between 2000 to 2022 in 202 countries (World Bank 2021;2024a).

Part two uses the QoG Standard TS dataset, WDI and scandal data. The WDI consists of cross-country time series data on indicators of economic development in 217 countries, available from 1960 to 2025, collected through administrative systems, international and national statistics agencies (WDI 2026). The dataset on scandals from Riberio et al., (2018) reports corruption scandals from 1987 to 2014 in Brazil, and 1989 to 2018 in Spain. The scandals were collected from publicly accessible web pages of the most popular news and media outlets in each country (Riberio et al., 2018: 991). In Riberio et al. (2018), they construct a complex network of observations on scandals to assess how individuals are involved using nodes for people involved in scandals, and ties that indicate whether two individuals were involved in the same scandal. Building on this foundational work, Pessa et al. (2025) uses this data to examine gender structures in political networks, assigning gender labels to each node. Each node (individual) in a corruption scandal represents either a politician, a public servant or a private actor. Due to ethical concerns and anonymity, the dataset only reveals gender but not occupation (Pessa et al., 2025: 2). I received the latest version from Professor Riberio, which is also the one used in Pessa et al.'s work. For each country, these CSV files contain columns for case (corruption scandal), year, gender_source (gender of individual), and gender_target (a pair of agents involved in the same case).

4.3.1 Dependent variable

The dependent variable (y) is corruption. In part one, corruption is operationalized using the variable *control of corruption* (wbgi_cce), which is a perception-based estimate, defining corruption as the exercise of public power for private gain by the World Bank. This variable is continuous with an indicator between -2, to +2,5. Negative values indicate weak control and more corruption, whereas positive and high values indicate stronger control and less corruption. The variable is available between 1996 to 2024 and covers 192 countries (Teorell et al., 2026: 1158; World Bank 2025a, 2025b; Kaufmann & Kray 2024).

In part two, the operationalization of corruption is threefold. My first dependent variable (DV1) examines the number of corruption scandals (num_cases) in a country-year. My second dependent variable (DV2) examines the number of women involved in a given scandal (total_females) in a country-year relative to the number of involved men. My third dependent variable (DV3) examines the probability of at least one woman being involved in a scandal (any_female) in a country-year. Both DV1 and DV2 are count variables, DV2 was generated into a dummy for the proportion of women, female_proportion (total_females/total_people *100). DV3 is also a dummy, coded as 1 if the total_females is larger than 0, meaning at least one woman have been involved in a scandal, and 0 if total_females equals zero women involved.

4.3.2 Independent variable

The independent variable (x) is women in top bureaucratic positions. For part one, the IV was merged from WWBI which contains gender-based demographic variables on female employment. The WWBI define females as a share of public paid employees by occupation: senior officials, divided into subcategories: professionals, technicians, clerks, elementary occupation workers and managers) based on ISCO 88 (World Bank 2021: 23). Managers, as a subcategory of female senior officials, refer to individuals with oversight and decision-making in planning, coordinating, evaluating activities in government and organizations, including chief executives, senior officials, managers and legislators (World Bank 2021: 24). I argue that managers constitute the closest available proxy for the number of women in top bureaucratic positions. Therefore, women in top bureaucratic positions are operationalized using the WWBI variable *public sector female employment, as a share of paid employment by occupation group*:

managers (female_manager_share). This variable shows the percentage of women in management positions in the public sector, available for 94 countries between 2000 to 2022.

Since this variable had to be merged into the QoG dataset, this resulted in an unbalanced panel since WWBI had more observations and countries available for this variable. My sample thus includes only country-years that have coverage in both the WWBI dataset and the QoG dataset. Also, since the IV is conducted in waves, some countries have several observations whereas others only have a few (see table A14 in appendix). Since FE models requires at least two observations for each country, I ran additional commands to assess the coverage and within-country variation, resulting in 94 countries with at least two observations, sufficient to conduct the analysis, and excludes countries with only one observation.

In part two, the IV assessing women in top bureaucratic positions only in Brazil and Spain required another set of data. The IV used in part one had barely any variation for Brazil, increasing the risk of not detecting any effect at all. Secondly, there was a lack of comparative national administrative data since there is no single definition that applies to both or available for the same period. The most valid indicator for both countries across the same time was the International Standard Classification of Occupations (ISCO) and International Labor Organization (ILO) variable: *female share of employment in senior and middle management (%)*. Collected through labor force and household surveys, this continuous variable describes the share of women employed in decision-making and management roles in government and public institutions, but also for large enterprises (WDI 2026). The main drawback with this variable is that it fails to differentiate between the public and private sector, and the differences in levels of responsibility of women in their respective managerial positions. Nevertheless, to my knowledge, this is the only variable capturing women in decision making roles in the public sphere (but also private), available for both Brazil (2002-2025) and Spain (2000-2025). Hence, I considered this IV the most applicable operationalization given limited data availability.

4.3.3 Moderator

To evaluate if the effect of women's representation in top bureaucratic positions on corruption is stronger in more democratic countries (H5), the moderator (z) democracy is operationalized

using the variable electoral democracy index (*vdem_polyarchy*)². This variable is an expert aggregate index of the level of representative democracy, electoral competition, freedom of expression, independent media, and civil rights. The index ranges from low (0) to high levels of democracy (1) and is available in 175 countries between 1946 to 2024 (Coppedge et al., 2025a, 2025b; Pemstein et al., 2025; Teorell et al., 2026:1247).

4.3.4 Control variables

To mitigate selection bias and spurious relationships, I incorporate a set of control variables for part one and two. Based on previous literature (Dollar et al., 2001; Swamy et al., 2001) suggesting that women in parliament have a negative effect on corruption, and potentially on women in the bureaucracy, I include *women in parliament* as a control. Women in parliament are operationalized using the variable the proportion of seats held by women in national parliaments in a single or lower chamber held by women (*wdi_wip*), available from 1997 to 2024 for 192 countries (Teorell et al., 2026: 1442; World Bank 2024b). Furthermore, prior work indicates corruption has detrimental effects on economic development and growth (Dye 2007; Rose-Ackerman & Palifka 2016). Hence, as a proxy for economic prosperity, I include *GDP per capita* (*wdi_gdpcapcon2015*), the gross domestic product divided by midyear population in constant 2015 U.S dollars, available from 1960 to 2024 and covers 191 countries (World Bank 2024; Teorell et al., 2026: 1320). Lastly, I control for *bureaucratic quality* on the level of corruption, operationalized using the variable government effectiveness (*wbgi_gee*), which measures the quality of the bureaucracy, the competence of civil servants and their independence from political pressure. This variable ranges from 1996 to 2024 and is available in 192 countries (Teorell et al., 2026: 1160; Kaufmann & Kraay 2024; World Bank 2025a, 2025b). Descriptive statistics for all DVs, IVs, and controls can be found in the appendix (see table AX).

² I also considered *p_polity2* and *vdem_libdem*. Running commands of correlation between the variables (*p_polity*, & *vdem_libdem*=0.8565, *vdem_libdem* & *vdem_polyarchy* = 0.9785, *p_polity2* & *vdem_polyarchy* 0.8984), show almost perfect correlation.

4.3.5 Ethics

This study relies on secondary panel data, cross-country surveys, and reported scandal data which only reveal the gender of individuals involved without any personally identifiable information being disclosed. Therefore, I suggest this study presents no major ethical concerns.

4.4 Limitations

There are limitations that require attention. Firstly, any measure of corruption, indirect or direct, is subject to measurement errors and biases (Piazza & Diaz 2024: 158). Hence, discussing the validity, reliability, and generalizability of this study is necessary. Validity refers to if a measure is valid based on repeated measurements and whether a variable measures the intended outcome (Adcock & Collier 2001: 530-531). For the DVs, corruption is measured by perceptions and by scandals. While the survey-based assessment of corruption may suffer from reduced validity due to response bias, the scandal-based measure offers higher validity by capturing a more observable indicator of corruption. The operationalization of the IVs required a degree of compromise, especially in part two since the proxy included public and private sector employees, potentially reducing validity by not fully capturing women in top bureaucratic positions. Furthermore, the slow-moving nature of the DVs and IVs in this study may in fact result in limited ability to detect variation in the regressions. Also, while the empirical strategy of the two-way FE strengthens the internal validity by controlling unobserved heterogeneity across countries and over time, reverse causality may still occur.

Reliability ensures that the collection of data is reliable and ensures that the procedure for the analysis can be replicated and yield equivalent results (King et al., 1994: 26). This study uses panel data and cross-country data which are comparable across various contexts. To ensure reliability for this study, I am transparent about the sources used, merging, and coding. Also, since corruption measures in two ways, both the validity and reliability would be strengthened if both analyses yield similar results, such as that women have a significant negative effect, both perception-wise and in actual engagement. In terms of generalizations, part one is advantageous since it provides broader cross-national coverage, implying that the findings may be more applicable across a larger set of countries. By contrast, part two is based solely on two countries, resulting in a smaller sample with less observations. Consequently, I am cautious

about drawing broader generalization may only be suitable for specific countries or countries like them.

Another concern is the combination of data. In part one, QoG and WWBI data was merged together, initially resulting in a unbalanced panel as the data was not perfectly aligned with the years they cover in data. This largely depends on that WWBI is recorded in waves due to that many countries lack the administrative and technological systems to regularly produce accurate data on public sector employment, challenging the ability to assess cross-national comparisons yearly (World Bank 2021: 12). As noted, the unbalanced panel required additional commands to ensure only matched observations across the datasets were included and resulted in 94 countries between 2000-2022. While considered sufficient, the generalizability decreases as some countries have more observations and other few and makes it harder for the FE framework to detect an effect, increasing the risk of selection bias. Likewise, part two combined three datasets, resulting in that the analysis could only be conducted between 2002-2014 to ensure overlapping observations, resulting in a small sample and losses in observations which reduced the model's statistical power.

Despite these methodological constraints, this study contributes to contemporary literature assessing the effect of women in top bureaucratic positions on both perception-based and behavioral measures of corruption, which is an approach not employed previously. To improve and strengthen confidence in my results, I conduct a set of robustness checks. Part one consists of three additional tests. Firstly, by examining yearly changes through first difference models. Secondly, using lags and delayed effects, showing the estimate for a variable from an earlier point in time. Thirdly, by changing the IV to the total share of female employees in the public sector (`wwbi_fspuemp`), to examine the effect without considering occupation. Part two also reports lags, and uses an additional model, Poisson FE (`xtpoisson`). Poisson FE is the most accurate model when the dependent variable is a count or nonnegative, reporting log-coefficients and incidence ratios (Stata n.d). Since DV3 is binary, I also run logistic regression. Logistic regression is the most appropriate modeling strategy for dichotomous dependent variables, calculating the probability for each observation having the value of 1 (Stathelp n.d-b). Lastly, both part one and two use robust and clustered standard errors to avoid heteroskedasticity and serial correlation.

5. Analysis

5.1 Part one

Part one is a cross-country perception analysis of women in top bureaucratic positions and corruption. This part tests if higher representation of women in top bureaucratic positions reduce corruption (H1) and whether the effect is stronger in more democratic countries (H2). Table 1 presents the bivariate OLS regression models³, Model 1 is a pooled OLS, Model 2 including year fixed effects (FE) and Model 3 including country FE and year FE. Negative coefficients mean that the control of corruption decreases and corruption worsens, whereas positive coefficients mean that the control of corruption increases and corruption lessens.

Table 1. Bivariate OLS regression

Dependent variable: Control of Corruption

	M1	M2	M3
Female managers in public sector (%)	-1.562*** (0.477)	-1.465** (-3.10)	0.0478 (0.36)
Country FE	No	No	Yes
Year FE	No	Yes	Yes
Constant	0.879*** (0.266)	0.442 (0.310)	0.442 (0.310)
N	664	664	664
adj. R ²	0.073	0.082	0.017

Robust standard errors in parenthesis at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset v2026 (Teorell et al., 2026) & WWBI (World Bank 2024).

Comment: Model 2-3 includes dummies for years, 2000 to 2022 (ref.2000).

In Model 1 all observations are pooled over the period, eliminating country-and time-specific effects. This model suggests that, across the 94 countries, a one-unit increase in female managers are on average associated with a 1.56 percentage point decrease in control of corruption between 2000 to 2022. Since the coefficient is statistically significant at the p<0.001

³ Cross-sectional OLS-model of female managers and control of corruption in a year, excluded and can be found in appendix see table A11 (the coefficient (β= -1.605), negative and insignificant effect.

level, this implies that the probability that this relationship is random is less than 0.1%. The model's explanatory power is considerably low since the adjusted R^2 report a 7.3% of the total variance of the DV. Substantively, the pooled model suggests that an increase in women in the bureaucracies is associated with worsening corruption in contrast to my expectations. Overall, model 1 cannot reject the null hypothesis.

Like Model 1, **Model 2** shows a negative and statistically significant coefficient ($\beta = -1.465$) at the $p < 0.001$ level, suggesting that controlling for global trends and time-specific effects across countries, a one-unit increase of female managers is on average associated with a decrease in the control of corruption. Removing only changes in years via year FE, this model still does not account for differences between countries, and the significant negative coefficient for female managers may be driven by select countries. Hence, model 2 shows no support for H1.

Model 3 is a two-way fixed effects (TWFE) model estimating country FE and year FE, limiting variation within a country and within each year of the period. The coefficient ($\beta = 0.0478$) is positive but insignificant and cannot be distinguished from zero. The reported within r square suggest a within variation of 0.0496, suggesting that changes in female managers explain approximately 4.96% of the within country variation in control of corruption over time. This model suggests that there is not robust evidence for an effect of female managers on corruption within-countries over time. In contrast to M1 and M2, this model does not provide evidence that women lead to worse corruption, but it also does not provide evidence that women improve corruption, either. Hence, model 3 cannot accept H1.

Table 2 reports full OLS regression models⁴. Across all models the coefficient for female managers is positive, yet only statistically significant in two.

⁴ CS-OLS model including interaction and controls can be found in appendix table A11.

Table 2. Full OLS-regression

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M5	M6
Female managers in public sector (%)	0.258 (0.136)	3.612*** (0.761)	0.708* (0.279)	0.113 (0.101)	0.404 (0.343)	0.400 (0.261)
Electoral democracy	0.315 (0.204)	5.868*** (0.752)	0.718* (0.327)	0.555*** (0.197)	0.570* (0.237)	0.728** (0.222)
Female managers* Electoral democracy		-6.195 *** (1.415)	-0.786 (0.480)		-0.532 (0.509)	-0.417 (0.392)
Women in parliament	-0.00114 (0.00321)		-0.00102 (0.00317)	-0.000904 (0.00223)		-0.000701 (0.00225)
GDP per capita	0.00000566*** (0.00000144)		0.00000540*** (0.00000145)	0.0000148* (0.00000681)		0.0000149* (0.00000670)
Bureaucratic quality	0.908*** (0.0536)		0.895*** (0.0537)	0.437*** (0.0749)		0.434*** (0.0746)
Country FE	No	No	No	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Constant	-0.149 (0.182)	-3.716 *** (0.521)	-0.408 (0.223)	-0.393* (0.193)	-0.106 (0.168)	-0.522* (0.206)
N	650	664	650	650	664	650
adj. R ²	0.920	0.551	0.921	0.249	0.033	0.251

Robust standard errors in parenthesis at 95% confidence intervals: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset v2026 (Teorell et al., 2026) & WWBI (World Bank 2024).

Comment: All models include dummies for years 2000 to 2022 (ref.2000).

Model 1 controls potential confounders by introducing control variables to the OLS with year FE pooled by country. Since the coefficient for female manages is insignificant, this model finds no effect on corruption across all countries over time, whereas controls such as GDP and bureaucratic quality are significant predictors for more control of corruption. Nonetheless, model 1 yields no support for H1.

Model 2 introduces the moderator of democracy to the OLS with year FE. The coefficient for female managers ($\beta = 3.612$), when holding democracy constant, reveals that a one-unit increase in female managers is associated with on average a 3.6 percentage point increase in control of corruption, and is statistically significant at the $p < 0.001$ level. Likewise, the interaction coefficient ($\beta = -6.195$) is significant and at the same significance level. However, since the coefficient is negative, this reveals that the positive impact (the increase) of female managers on corruption decreases as electoral democracy increases. Figure 1 demonstrates this relationship, highlight the marginal effect of female managers on the control of corruption, depending on the level of electoral democracy. When democracy equals 1, the effect is approximately -2.6 and statistically significant, implying that more female managers is associated with less control (more corruption), whereas where democracy is low, or entirely authoritarian ($=0$), the effect is 3.6, indicating that the impact of more female managers is associated with more control (less corruption). The effect of female managers becomes null around 0.6 on the electoral democracy index where the line crosses zero. Thus, the effect seems conditional as the influence of more female managers on control of corruption may be greater in contexts where electoral democracy is weaker. Therefore, Model 2 essentially displays support for H1 but not for H2.

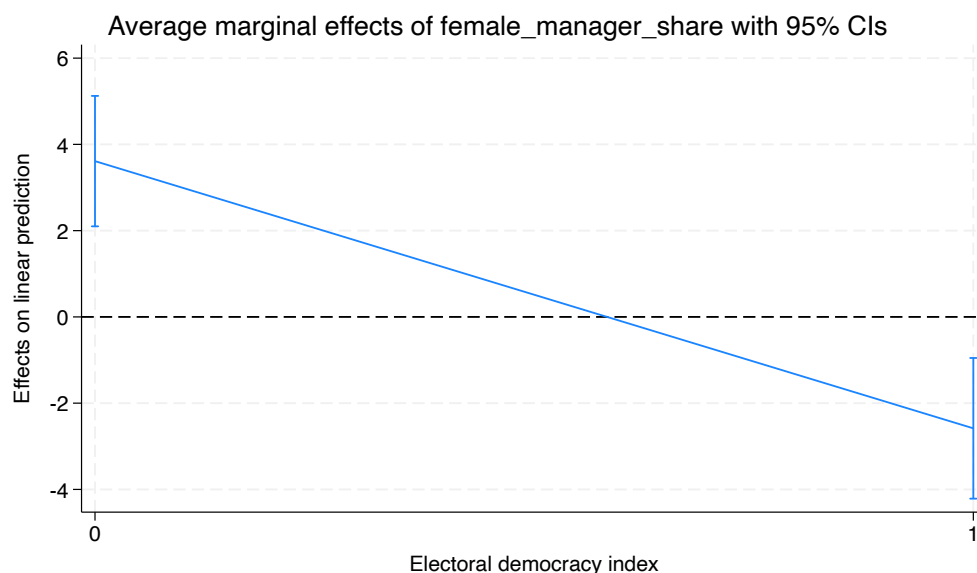


Figure 1. Marginal effects

Model 3 introduces a moderator and confounders to the OLS with country and year FE. Like model 2, when holding democracy constant, the coefficient for female managers ($\beta = 0.708$) is positive and significant at the $p < 0.05$ level, supporting earlier findings that female managers seem to exhibit greater influence over the control of corruption when the level of electoral democracy is lower across all countries over the period. Yet, when including the controls, the coefficient for the interaction becomes smaller and insignificant ($\beta = -0.786$). Hence, the preliminary evidence from model 2 suggests that increases in female managers are associated with less control of corruption in more democratic countries may be driven by omitted variable bias since the significance of the interaction coefficient does not hold when including controls. Like model 1, controls such as GDP and bureaucratic quality remain significant predictors for more control of corruption. Across all models in table 2, model 3 indicates most explanatory power since the adjusted r square reports that the model explains about 92.1 % of the total variance of the dependent variables. Nevertheless, model 3 cannot support H2.

Model 4 assesses two-way FE including controls, holding year and country factors constant. Like previous TWFE models in table 1, the coefficient ($\beta = 0.113$) for female is insignificant, implying that within-countries over time the effect of female managers on corruption cannot be distinguished from zero and is probably driven by other explanatory factors, denying H1. Nevertheless, electoral democracy, GDP per capita, and bureaucratic quality remain significant predictors for greater control of corruption. Similarly, **model 5** of TWFE including the interaction yet excluding controls, both the coefficient of female managers and interaction are insignificant, rejecting both hypotheses, and further suggests that the relationship may only hold between but not within-countries.

Lastly, **model 6** examines TWFE with interaction and controls, and exhibits almost equivalent results to model 5. Including both country and year FE, the effect of female managers on control of corruption when both holding democracy constant ($\beta = -0.400$), and the interaction ($\beta = -0.417$), is negative and insignificant, whereas the controls (electoral democracy, GDP, and bureaucratic quality) are significant predictors for more control of corruption. In model 6, the within-variation suggests that the change in female managers explains about 28.1% of the within country variation in control of corruption over time. This model shows an interesting pattern in terms of control of corruption: coefficients for year fixed effects show that, compared

to the reference year (2000), in 2004, 2005 and 2007 the control of corruption was negative and statically significant, and from 2012 through 2019, the coefficients show an increasing negative trend which reaches its largest negative effect ($\beta = -0.259^{**}$) in 2019, highlighting how control of corruption has deteriorated over the period for all countries. Nevertheless, investigating within-country variation each year, there is a lack of robust evidence for female managers effect on corruption, and is more likely to be driven by between country variation, as the coefficient for female managers in model 2 and 3 predicts. Thus, model 6 also rejects H1 and H2.

5.1.1 Robustness checks

To test the robustness of the findings of the original analysis, I conducted three robustness checks, first-difference (FD) models, lags, and change of IV. Evidently, running these additional tests suggests that the effect of female managers on corruption appears sensitive to model specification. Firstly, I explored FD models (see A12 in appendix) assessing year to year changes of the effect of female managers on control of corruption. The bivariate regression shows that the yearly changes from one year of increases in female managers effect on corruption compared to previous year yield positive but insignificant effects. Both FD models of OLS with only year FE ($\beta = 0.135$) and OLS including controls ($\beta = 0.149$), suggest positive and significant effect as the $p < 0.05$ level and $p < 0.01$ level respectively. Yet, the full models including interaction and confounders, exhibit similar results as the original model, showing no robust evidence for female managers effect on corruption. Given that FD models reveal yearly changes, this approach may be less suitable when both the DV and IV vary slowly.

Furthermore, I conducted two robustness tests to assess delayed effects using one-year (see A3 in appendix) and five-year lags (see A4 in appendix). Across the models, the coefficient for female managers and the coefficient for the interaction of female managers and electoral democracy is highly similar to the models in table 2. The models only detect a significant coefficient between countries in the OLS models with year FE but not in the TWFE models assessing within countries over time. Hence, like table 2, the models assessing a potential delayed effect show that an increase of female managers in the previous year (L) or previous five years (L5) is associated with more control of corruption but only in countries that are less democratic and not in more democratic countries. Thus, introducing lags supports the previous findings in the full model in table 2, suggesting the effect on female managers on corruption

seems contextual and sensitive to choice of models, only detecting a significant effect across countries but not within countries over time.

Lastly, I ran an additional test by changing the IV to *females, as a share of public paid employees* (wwbi_fspuemp), to assess whether the relationship depends on how women's representation is measured. Surprisingly, changing the IV to all public sector female employees (see A5 in appendix) points in an opposite direction. The statistically significant coefficient for the IV suggests that increases in female public sector employees are on average associated with more control of corruption in more democratic countries, both between and within-countries. However, in the full model, including interaction, confounders, year and country FE, the coefficient for female public sector employees is insignificant, implying the findings are not very robust within countries. Nonetheless, the divergent findings changing the IV might reflect substantial differences between female elite level representation and the overall female workforce within the bureaucracy since only the latter seems to reduce corruption in more democratic countries, questioning the robustness of the findings in the original analysis.

5.2 Part two

Part two examines female bureaucrats' engagement in corruption, assessing the relationship between women in top bureaucratic positions and corruption using data on corruption scandals in Brazil and Spain. This section tests if the effect on women in top bureaucratic positions reduce corruption (H3), decrease female engagement in corruption (H4), and whether the association between more women in top bureaucratic positions and lower female involvement in corruption is greater in low-corruption contexts than high-corruption contexts (H5). Table 3 display descriptive statistics of the number of scandals and female involvement in percent by country and year. Overall, Spain has significantly more scandals compared to Brazil, and female involvement is often between 15-18%. In contrast, most scandals in Brazil consist solely of men, and for several years there are scandals without any female participation at all. These descriptive findings are notable since Spain may contribute with substantially more observations than Brazil, which may cause disproportional influence in the estimates, and adhere more specifically to Spain rather than generalizability across both countries.

Table 3. Number of scandals and female involvement by Country and Year

Year	Brazil N scandals	Brazil Total N	Brazil Female (%)	Spain N scandals	Spain Total N	Spain Female (%)
1987	2	28	0	-	-	-
1988	1	12	0	-	-	-
1989	-	-	-	1	1640	17
1990	-	-	-	1	2	100
1991	1	12	33	1	2	0
1992	1	220	18	1	72	36
1993	2	48	0	1	2	0
1994	1	4	0	1	30	20
1995	1	4	0	2	4074	1
1996	-	-	-	-	-	-
1997	4	332	15	2	4	0
1998	3	236	13	6	426	26
1999	2	52	0	5	614	4
2000	1	84	0	11	5280	6
2001	2	152	23	6	1192	5
2002	3	244	17	7	1082	10
2003	2	544	9	26	3420	17
2004	4	152	4	27	2830	15
2005	5	4684	2	25	3790	15
2006	3	3148	5	24	2784	18
2007	5	44	7	34	3534	17
2008	4	152	43	42	2432	14
2009	3	1256	30	36	9108	17
2010	3	256	4	20	3204	12
2011	6	124	3	25	1844	21
2012	3	906	38	22	6578	17
2013	1	264	17	10	346	17
2014	2	1246	9	10	1402	11
2015	-	-	-	9	268	26
2016	-	-	-	7	300	36
2017	-	-	-	3	314	1
2018	-	-	-	2	4	100

Part two proceeds by conducting country-year analysis on female managers effect on number of scandals (DV1) and number of women involved in scandals (DV2). As mentioned in the methodology, to conduct a comparable analysis, and due to data unavailability, the panel regressions with country-year were restricted to the period 2002-2014. This resulted in a small N which lowers the statistical power of the models. Aware of this limitation, the findings of these models should be interpreted cautiously. The models are underpowered, and their ability to detect robust associations between variables is therefore weak. The analysis begins with models looking at each country separately (model 1-4), while model 5 and onwards assess both countries together, and control for country differences with dummies and controls. Table 4 presents a summary of the effect of female managers on corruption scandals (DV1).

Table 4. Summary table for DVI

Dependent variable: Corruption scandals							
	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6	M7
Female senior managers	0.729*** (0.141)	-0.285 (0.695)	-0.684 (.)	0.319 (.)	-0.294 (0.671)	-1.764 (2.232)	-0.138*** (0.003)
Democracy						-96.441 (317.997)	-33.948*** (1.060)
Women in parliament						-2.262 (1.757)	-0.259*** (0.001)
GDP per capita						0.008* (0.003)	0.000*** (0.000)
Bureaucratic quality						-56.917 (28.165)	-3.642*** (0.064)
Country FE	No	No	No	No	Yes	Yes	-
Year FE	No	No	Yes	Yes	Yes	Yes	-
Poisson FE	No	No	No	No	No	No	Yes
Constant	-22.895*** (4.863)	30.449 (16.657)	26.967 (.)	1.207 (.)	4.356 (24.288)	98.137 (326.422)	
Observations	12	13	12	13	25	25	25
Adjusted R ²	0.225	-0.076	.	.	0.658	0.866	.

Robust standard errors in parentheses at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Year FE included in all models except 1-2 (ref.2002). Country FE included in FE and Poisson FE in Model 7.

Model 1 (Brazil) and **Model 2** (Spain) test the relationship between female managers and corruption scandals in each country. Brazil exhibits a statistically significant and positive coefficient ($\beta=0.729$) at the $p<0.001$ level, suggesting that the effect of an increase of female managers would on average be associated with an increase in scandals. Spain shows a negative coefficient ($\beta=-0.285$), suggesting that scandals would decrease, yet insignificant effect. While Spain provides no support for H3, Brazil suggests the opposite. **Model 3** (Brazil) and **Model 4** (Spain) assess the relationship including year FE, separately for each country. Controlling for time-specific effects, the coefficient ($\beta=-0.684$) for Brazil becomes insignificant and negative, increasing the likelihood that previous findings are spurious, whereas for Spain the coefficient ($\beta=-0.319$) is positive but insignificant. **Model 5** considers Brazil and Spain together and uses OLS with year FE and country FE. This model tests the relationship between female managers

and corruption, holding time-variant and time-invariant differences constant between the two countries, and does not find any statistically significant effect. **Model 6** also employs country and year fixed effects while including controls, and this model further confirms the null effect.

Lastly, **Model 7** is a robustness check, using a Poisson regression instead of a linear regression. This model is a more precise estimate of the relationship given the count nature of the dependent variable. Poisson regressions are the most appropriate way to model count data, but the interpretation of Poisson models is less straightforward than that of linear regressions, which still perform well in modeling count data and are therefore frequently used with count-style dependent variables due to this advantage. Therefore, Poisson is an additional model to further explore the relationship between my IV and DV1, complementing the main modeling strategy for DV1. **Model 7** suggest that an increase in female managers would be associated with approximately 12.9%⁵ lower expected frequency of corruption scandals, and statistically significant at the $p < 0.001$ level. Overall, building on a small N, the findings in model 1-6 suggest that there is no robust evidence for H3. Nevertheless, model 7 using Poisson FE, which may be a more precise fit of the data, finds a significant effect, highlighting the sensitivity for model specification.

Table 5 shows a summary of the effect of female managers on female involvement in scandals (DV2). Both **Model 1** (Brazil) and **Model 2** (Spain) find positive coefficients for female managers, initially suggesting that an increase of female managers would on average be associated with an increase of female involvement in scandals. However, since both coefficients are not statistically significant, there is no support for H4. Introducing year FE to each country separately, **Model 3** (Brazil) and **Model 4** (Spain) likewise suggest insignificant effects, although the coefficient for Spain turned negative. **Model 5**, which includes both Brazil and Spain, and restricts identification to within-country and within-year, also suggests no statically significant relationship. **Model 6**, controlling time-specific effects, country characteristics, and confounders, also yields a negative yet insignificant effect. Given the nature of the coefficients across all models in table 5, there is no robust evidence supporting female managers effect on

⁵ The coefficient ($\beta = 0.138$) is exponentiated and converted to incidence rate ratio.

female involvement in scandals, even when pooling the observation together, and holding country and year FE constant. Thus, both H4 and H5 must be rejected.

Table 5. Summary table for DV2

Dependent variable: Female involvement in scandals (%)

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6
Female senior managers	0.282 (1.072)	0.185 (0.449)	6.031 (.)	-0.812 (.)	-0.482 (0.532)	-0.890 (1.320)
Democracy						205.837 (203.623)
Women in parliament						1.453 (1.983)
GDP per capita						-0.003 (0.004)
Bureaucratic quality						9.964 (22.387)
Country FE	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Constant	-0.264 (39.772)	16.507 (10.203)	-210.358 (.)	33.765 (.)	19.773 (17.885)	-133.363 (212.322)
Observations	12	13	12	13	25	25
Adjusted R^2	-0.096	-0.078	.	.	0.574	0.386

Robust standard errors in parentheses at 95% confidence interval: *** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.05$.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Year FE included in all models except 1-2 (ref.2002). Country FE included in FE.

While DV1 (table 4) and DV2 (table 5) use aggregated country-year as unit of analysis, DV3 evaluates at case-level how the presence of female managers affects the probability for that a given scandal case has involved a woman or not. Assessing the effect at case level allows for examining multiple scandals in a country-year, increasing the sample size significantly ($n=349$) and enhancing the analysis of statistical power. Since DV3 is dichotomous (1= any female involved in a scandal), and the analysis focuses on scandals, the panel data set up could not be performed since the same scandal does not appear twice. Acknowledging this, models with DV3 use linear probability model (LPM). Table 6 presents a summary of the estimates.

Table 6. Summary table of DV3

Dependent variable: Probability for any female involved in scandals

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6
Female senior managers	-0.208** (0.067)	-0.007 (0.008)	0.114 (0.363)	-0.033 (0.025)	-0.044* (0.022)	0.023 (0.082)
Democracy						15.608 (11.101)
Women in parliament						0.187** (0.057)
GDP per capita						-0.000 (0.000)
Bureaucratic quality						1.874 (1.033)
Country FE	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Constant	8.076** (2.441)	0.678*** (0.186)	-3.661 (12.915)	1.321* (0.596)	2.008** (0.710)	-13.964 (11.508)
Observations	41	308	41	308	349	349
Adjusted R ²	0.177	-0.000	0.325	-0.002	0.002	0.006

Robust standard errors in parenthesis at 95% confidence interval: ***p<0.001, **p<0.01, *p<0.05.

Source: Scandal data 2002-2014 (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Linear probability models (LPM) between 2002-2014. Year FE included in models 3-6, and Country FE in model 5-6.

Like tables 4 and 5, I first employ bivariate regression for each country. **Model 1** of Brazil suggest a positive significant effect at the $p<0.001$ level, finding that a one-unit increase in female managers decreases the likelihood for that a female has been involved in a scandal by 20.8 percentage points. In **Model 2** of Spain, the coefficient is negative and insignificant, detecting no effect at all. **Model 3** (Brazil) and **Model 4** (Spain) in table 6 show separate bivariate OLS regressions with year FE. Neither model show significant coefficients for female managers, finding that a one-unit increase in female managers has essentially no effect on the probability for that any female has been involved in a scandal. While Spain had no significant coefficients for years in dummies, in Brazil, during 2008 (0.574*), 2009 (0.644*) and 2012 (0.598*) finds that the scandals had significant higher probability of involving women compared to the reference year. Nevertheless, for models 1-4, the examined period between

2002 to 2014, Spain exhibits 308 observations whereas Brazil only has 41. Hence, in the following models 5-6, these findings should be interpreted with caution since they may be driven more by Spain than by Brazil.

Model 5 examines the association controlling for both year FE and country FE, removing all distinct country characteristics between Brazil and Spain. This model shows that a one unit increase in the share of female managers is on average associated with a 4.4 percentage point lower probability that a given scandal involved women, for both countries, significant at the $p < 0.05$ level, supporting H5. However, as mentioned, these findings are probably affected by the unbalanced number of observations since most observations originate from Spain. Lastly, **Model 6** enhance **Model 5** by including controls. Holding time-invariant, and time-variant factors as well as confounders constant, the previous significant evidence for female managers effect on the probability for women's inclusion in scandals disappears. Hence, the findings in **Model 5** of the lower probability for women's involvement in scandals is likely affected by omitted variable bias, indicating that findings should not be interpreted substantially since **Model 6** prove the evidence is not robust. Overall, none of the models in table 6 identifies robust evidence for H4 nor H5.

5.2.1 Robustness checks

To evaluate the robustness of the findings in table 4 (DV1) and table 5 (DV2), I conducted one-year and five-year lags. The inclusion of lags resulted in significant reductions in observations, especially in the five-year lags, indicating these findings should be taken with considerable caution. Overall, for DV1, the effect of a one-year lag (see models in table A6 in appendix) and the five-year lag (see models in table A7 in appendix) supports the idea of a delayed effect, suggesting that increases in the proportion of female managers decreases the number of scandals between Brazil and Spain, holding country FE and year FE constant. While the original analysis exhibited no support for H3, the lag models, especially the five-year, at least consider that the effect may simply be delayed. For DV2, the effect of a one-year lag (see models in table A8 in appendix) revealed almost equivalent findings to the original analysis. Yet, the effect of a five-year lag (see models in table A9 in appendix) deviates from the original analysis showing positive and insignificant coefficients, suggesting the relationship may not necessarily be explained by delayed effects but by other confounding variables not explored in this analysis.

Thus, assessing delayed effects on DV2, H4 is only supported in the models of a one-year but not five-year lag.

The robustness of the findings in table 6 (DV3) is examined using logistic regression with margins (see table A10 in appendix), ensuring that the coefficients could be interpreted similarly to the LPM's. Overall, the marginal effects in the logistic regression are generally consistent with the coefficients in the LPM models, exhibiting no greater differences. The coefficient for female managers in Model 1 for Brazil remains significant and negative, whereas for model 5, the coefficient for female managers, at a 95% confidence show almost significance negative effect ($\beta = -0.187^+$). Nevertheless, since the estimates in the logistic regression are almost equivalent to the original model, this additional test finds no robust evidence for that the probability for women's involvement in each scandal is affected by increased presence of female bureaucrats.

6. Discussion

To what extent does women's representation in top bureaucratic positions affect corruption?

Empirically, like politicians, bureaucrats are also very involved with corruption (Alhassan-Alolo 2007; Brierley & Pereira 2023; Carreras et al., 2024). While politicians may exert more access, power and resources to corruption, the bureaucracy, constituting a central element of public service delivery, may either operate impartially or become corruptible. In line with the literature, part one examined the broader relationship between women in the bureaucracy and corruption using perception-based measured. Unlike the preliminary work by Dollar et al., and Swamy et al., but consistent with the findings of Stensöta et al., this study shows that women's influence on corruption may not necessarily hold outside the political realm. Although Stensöta et al., focused on women in the bureaucracy more broadly, accounting for the occupational position, women in top bureaucratic positions, does not yield significant effects on corruption. Higher representation of these women and lower corruption appears to be driven more by differences between countries than within and is likely driven by other confounding factors. Also, given the study's methodological constraints, the findings were interpreted cautiously and did not provide robust support for H1.

The theoretical mechanisms commonly used to assess gender differences in corruption may to some extents have overestimated the influence of gender, as this study suggests that the effect is not generalizable to women in the bureaucracy and may instead be driven more by institutional factors. This aligns with prior work suggesting that the relationship between gender and corruption is highly contextual (DiRizzeno 2019; Esaray & Chirillo 2013) and that it only holds in well-functioning democracies with liberal institutions (Sung 2003). Hence, in response, this study also examined the conditional effect of democracy. Consistent with the literature (e.g. Sung 2003, Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2019), this study also finds that the relationship is contextual, however, in the opposite direction. When electoral democracy increases, the positive impact of women in top bureaucratic positions on corruption decreases. This implies that greater presence of these women is only likely to reduce corruption in less democratic countries but not in more democratic countries.

The implications of these findings suggest that the independent role that presence of women in these occupations have on corruption may not be as big or important when democracy is already high, whereas in less democratic countries with more corruption, the presence of women is likely to have more on an effect. Since prior work suggest young democracies tend to have more corruption (Bäck & Hadenius 2009), although speculative, it could be that less democratic countries may strategically incorporate more women in these positions as a legitimating strategy to signal they are committing to democratic norms, and like authoritarian regimes, use women as a form of window washing by saying having more women they are also reducing corruption. On the other hand, these findings could also reflect arguments by Bauhr and Charron suggesting that even in well-established democracies elite women may become more susceptible to corruption over time. Given that women in top bureaucratic positions are underrepresented in both more democratic and less democratic countries, they may, like female mayors, similarly adapt to corrupt networks rather than challenging them. However, as noted by Esaray and Schwindt-Bayer, since corruption is stigmatized in well-functioning democracies these women in more democratic settings may be aware of corrupt practices, but they refrain from acting against them due to the institutional pressures or career-related constraints suggested by Bishu et al. and Heilman. Nevertheless, these findings do not necessarily mean that women in top bureaucratic positions in well-functioning democracies are more corrupt,

rather it implies that there may be other stronger factors and mechanisms regulating corruption than simply the presence of these women. Evidently, this study fails to find robust support for H2.

Overall, the perceptions-based approach suggests the extent to which women's representation in top bureaucratic positions affect corruption is somewhat ambiguous since this study shows that there is not a strong relationship. Moreover, this study suggests context play a pivotal role in shaping the relationship between gender and corruption, and that the role of women in the bureaucracy may operate differently from the legislature. The absence of significant effects in this study is still theoretically meaningful as it speaks for the relevance of more research on this topic. In line with Stensöta et al., 2015, given that bureaucrats lack political authority, this factor alone may in itself affect bureaucrats' opportunities to engage or deter corruption. Future research should therefore examine these women's attitudes and their decision-making related to corruption across contexts, which is an area that remains relatively underexplored.

How does women in top bureaucratic positions influence women's involvement in corruption?

The literature on gender and corruption suggest that women's lower engagement in corruption may reflect theories of the fairer sex hypothesis (Dollar et al., 2001), risk-aversion (Esaray & Chirillo 2013; Esaray & Schwindt-Bayer 2017) or outsider-status (Goetz 2009; Barnes & Beaulieu 2018). Unlike most work on gender and corruption which relies on macro-level corruption measures, part two explored the effect of women in top bureaucratic positions on observable indicators of corruption and women's engagement using scandal data from Brazil and Spain. At glance, the descriptive statistics show that women's engagement in scandals in both Brazil and Spain is low. Hence, descriptively aligning with previous literature suggesting that women are generally less involved in corruption (Goez 2009) and most corrupt networks involve men (Stockemer & Sundström 2019). On the other hand, this study showed no strong relationship between women in top bureaucratic roles and decreasing number of scandals, female involvement in scandals nor the probability for female involvement. The pattern of insignificant effects suggests that similar to the findings in part one, women in top bureaucratic positions do not systematically reduce corruption and are neither less nor more likely to be involved in corruption, leading to the rejection of H3-H5. These findings challenge the existing theories in the literature concerning women's honesty, risk-aversion and exclusion since these

mechanisms may not necessarily generalize to women at the upper ranks of the bureaucracy. Furthermore, this study challenge both the idea of more women is associated with less corruption, and that women should be cleaner in more well-functioning democracies with less corruption. Contrary to Esaray and Schwindt-Bayer, this study does not find enough support for that more democratic countries with effective institutions and less corruption, in this case Spain, women would be more risk-averse and less corrupt. Also, while Bauhr and Charron could see that reelected female mayors did not decrease corruption, suggesting that elite women may become more corrupt over time, this study of “*elite*” women in the bureaucracy fails to provide any nuance to this argument. Essentially, there is not enough evidence for each country separately nor together to confirm how women’s engagement in corruption in low corrupt contexts, like Spain, deviates from women in high corruption contexts, like Brazil.

The broader implication of this study is that women’s involvement in corruption may be structurally driven rather than by gender alone, and that increased presence of women in the bureaucracy does not seems to operate as an anti-corruption mechanism. Given that part two was subject to methodological limitations, including a relatively small and skewed sample, the findings were interpreted cautiously, and future academia would benefit from extending the scope by increasing the number of comparable observations and countries included. This study was designed to capture patterns of women’s involvement in corruption, yet it did not directly measure underlying incentives. Hence, the insignificant findings point to that there may be other explanatory factors, not captured in this study, that better explain female bureaucrats’ engagement in corruption. Prior research has highlighted other individual-level determinants of bureaucratic corruption, such as salary levels (Graeff et al., 2025), turnover rates (De Vries & Sobis 2016), and politicization (Meyer-Sahling et al., 2018; Oliveros & Schuster 2018). Although these factors lie beyond the scope of the present study, they exhibit important avenues for future research and may provide stronger explanations for women’s involvement in corruption, rather than the effect of more women occupying top positions within the bureaucracy. How women in top bureaucratic positions influence women’s involvement in corruption is not clearcut as the findings of this study suggest that gender is not a consistent predictor of corruption in the elite bureaucratic settings, rather contextual or institutional factors seem more likely to be predictors shaping corrupt behavior in the bureaucracy.

7. Conclusion

Like female elected officials, women in top positions within the bureaucracy remain a minority worldwide. Despite their authority and influence over core bureaucratic functions, this study finds insufficient evidence that these women reduce corruption to the same extent as women in legislative institutions. This study suggests that gender and corruption are contextual since there are generally no strong findings for an effect on bureaucracy, except under certain circumstances in less democratic countries. However, even this evidence remains weak and not especially robust. Furthermore, comparing high-corrupt contexts, such as Brazil, with low corruption contexts, such as Spain, this study does not provide evidence for robust differences of the effect of women's presence in these positions on women's involvement in corruption. Thus, in accordance with scholars such as Alhassan-Alolo, Alatas et al., and Stensöta et al., this study suggests that increases of women in the public sector is simply not an antidote for corruption

Consequently, this thesis positions itself between existing perspectives in the literature on gender and corruption since it can neither confirm that these women in the bureaucracy reduce corruption nor conclude that they necessarily increase it. Also, the analysis reveals that these women do not have a distinct effect on women's involvement in corruption. However, this study still shows that there are women that engage in corruption and even if they are few, this phenomenon remains important to examine further because we do not know the mechanisms behind driving their incentives. I contend that women in top bureaucratic positions are not immune to corruption, and may, under certain conditions be particularly vulnerable to it. Unlike elected officials, these women are not held accountable to voters, and because they are appointed or recruited through merit, rather than elected, highly politicized environments may increase pressures for them to participate in existing corrupt practices. However, corrupt behavior among these women may remain less visible than among female politicians. At the same time, in accordance with the literature noting women's underrepresentation and vulnerability in these positions, may further increase their reluctance to engage in corruption, to maintain legitimacy and job security. Drawing on these speculations, the factors shaping female bureaucrats' propensity to engage in corruption remain insufficiently explored and deserves further attention.

The real-world implication of this study suggests that the relationship between gender and corruption in the bureaucracy may not be analyzed under the same assumptions as the legislative sphere, and policymakers must consider this when designing policies. For example, the promotion of gender quotas as an anti-corruption tool should be avoided in the context of the bureaucratic realm, as increasing women's participation is unlikely to reduce corruption. Rather than placing it as a strategy to reduce corruption, policymakers should advocate gender quotas as a policy to strengthen gender equality in the bureaucracy, especially since women in top bureaucratic positions are underrepresented. Moreover, since gender has no clear effect on corruption within the bureaucracy, policymakers should place greater emphasis on institutional factors and monitoring of these structures. Anti-corruption policies should aim at evaluating more closely the conditions that facilitate corruption. This is particularly important given that corruption within the bureaucracy may be more difficult to detect and remains more hidden as bureaucrats are not subject to the same level of accountability and scrutiny as politicians.

This thesis also presents avenues for future research on gender and corruption. While this study relied on observational data, future studies could expand the analysis by including a larger set of countries and observations, using experimental or field designs. Future work could also examine other areas of the bureaucracy and different occupational roles held by women and compare with women in similar roles within the private sector. Future research could also explore whether a critical mass of women in top bureaucratic positions is required for an observable effect, meaning that is possible a relationship may only emerge once representation exceeds a certain threshold. Furthermore, future avenues would benefit from combining quantitative analysis with qualitative approaches, to better understand the factors shaping these women's involvement in corruption. Fieldwork and interviews could provide deeper insights into the behavioral patterns of these women, whereas more comprehensive data on public administration and corruption would provide greater generalizability across contexts, even if the latter remains challenging to obtain.

To the best of my knowledge, this study is the first to explore how women in top bureaucratic positions affect corruption. While the findings are ambiguous and not fully conclusive, this study provides some important takeaways. Overall, this thesis does not find robust evidence for that higher participation of women in the bureaucracy is associated with less corruption,

questioning the theoretical expectations that increasing women's presence should clean up government by reducing corruption. Moreover, this thesis highlights the need for more research on women outside the political sphere, especially those in decision-making roles, as they may be shaped by corruption in distinct ways that do not necessarily pertain to women within the legislature. This research also holds policy relevance because policy makers can benefit from information on efforts to reduce corruption within the bureaucracy while simultaneously promoting policies on gender equality and women's career advancements.

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Appendix

Table A 1 Descriptive statistics part one

Variables	Obs	Mean	Std.dev	Min	Max
Public sector female employees by occupation: managers (%)	682	0.389	0.169	0.048	1
Control of Corruption	664	0.272	0.961	-1.060	2.302
Electoral democracy	682	0.695	0.214	0.1	0.921
Women in seats in national parliament (%)	666	21.430	10.875	0.917	63.75
GDP per capita	682	17557.21	21873.17	243.0767	112417.9
Bureaucratic quality	663	0.336	0.879	-2.081	2.273

Table A 2 Descriptive statistics part two

Variables	Obs	Mean	Std.dev	Min	Max
Female senior and middle managers	25	29.63	7.03	15.84	38.07
Corruption scandals (count)	26	13.54	12.63	1	42
Corruption scandals including women (proportion)	26	15.23	8.44	0.819	32.92
Women in a given scandal – case level (%)	26	1	0	0	1
Electoral democracy	26	0.875	0.010	0.856	0.893
Women in seats in national parliament (%)	26	21.92	13.79	6.24	39.71
GDP per capita	26	16999.77	9025.65	6941.44	27338.8
Bureaucratic quality	26	0.5094679	0.688	-0.387	1.793

Table A 3 Robustnesscheck one-year lag

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M5	M6
Public sector female employees by occupation: managers (%)	0.176 (0.180)	4.535*** (1.163)	1.062* (0.403)	-0.117 (0.143)	-0.473 (0.635)	-0.0404 (0.588)
Electoral democracy	0.286 (0.228)	6.881*** (1.071)	1.022* (0.444)	0.753** (0.300)	0.616 (0.316)	0.789* (0.377)
Female managers*Electoral democracy		-7.249 *** (1.875)	-1.399* (0.643)		0.286 (0.883)	-0.0981 (0.798)
Women in parliament (%)	-0.00276 (0.00387)		-0.00252 (0.00374)	-0.000424 (0.00251)		-0.000371 (0.00257)
GDP per capita	0.00000546*** (0.00000148)		0.00000507** (0.00000145)	0.00000890 (0.00000746)		0.00000887 (0.00000742)
Bureaucratic quality	0.946*** (0.0568)		0.924*** (0.0569)	0.297*** (0.0828)		0.298*** (0.0838)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	No	No	No	Yes	Yes	Yes
Constant	-0.216 (0.173)	-4.650 *** (0.766)	-0.699* (0.293)	-0.438 (0.309)	- 0.0340 (0.253)	-0.467 (0.374)
N	451	466	451	451	466	451
adj. R ²	0.931	0.549	0.933	0.163	0.080	0.161

Robust standard errors in parenthesis at 95% confidence intervals: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QOG Standard TS Dataset Version Jan26 (Teorell et al., 2026)

Comment: One year lag (L.) All models include dummies for years 2000 to 2022 (ref.2000)

Table A 4 Robustnesscheck five-year lag

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M5	M6
Public sector female employees by occupation: managers (%)	0.170 (0.262)	4.556** (1.314)	1.323*** (0.554)	0.0767 (0.169)	0.380 (0.523)	0.0299 (0.518)
Electoral democracy	0.514 (0.309)	6.930*** (1.178)	1.463* (0.569)	0.610 (0.506)	0.143 (0.484)	0.591 (0.604)
Female managers*Electoral democracy		-7.830 *** (1.862)	-1.807* (0.812)		-0.518 (0.738)	0.0606 (0.747)
Women in parliament (%)	-0.00467 (0.00552)		-0.00471 (0.00530)	-0.000125 (0.00423)		-0.00121 (0.00435)
GDP per capita	0.00000630*** (0.00000204)		0.00000592*** (0.00000200)	-0.00000968 (0.00000901)		-0.00000962 (0.00000909)
Bureaucratic quality	0.860*** (0.0758)		0.827*** (0.0754)	-0.0124 (0.0966)		-0.0125 (0.0962)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	No	No	No	Yes	Yes	Yes
Constant	-0.234 (0.291)	-4.671*** (0.925)	-0.917* (0.420)	-0.0517 (0.419)	0.154 (0.341)	-0.0402 (0.455)
N	311	323	311	311	323	311
adj. R ²	0.899	0.568	0.903	0.089	0.057	0.086

Robust standard errors in parenthesis at 95% confidence intervals: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026)

Comment: Five-year lag (L5.) All models include dummies for years 2000 to 2022 (ref.2000).

Table A 5 Full OLS regression changed IV

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M5	M6	M7
Female managers in public sector (%)	-0.372 (0.354)	-0.0688 (0.302)	-6.121*** (1.319)	-1.252* (0.539)	0.366 (0.427)	-0.965 (0.523)	-0.641 (0.891)
Electoral democracy	0.344 (0.375)	0.226 (0.258)	-2.782* (1.066)	-0.561 (0.508)	0.516* (0.201)	-0.565 (0.461)	-0.144 (0.474)
Females as a share of public paid employees * Electoral democracy			11.34*** (1.993)	1.896* (0.838)		1.931* (0.955)	1.363 (0.887)
Women in parliament (%)	-0.00778 (0.00430)	-0.000769 (0.00328)		-0.00109 (0.00321)	-0.00108 (0.00222)		-0.00105 (0.00224)
GDP per capita	0.00000641** (0.00000165)	0.00000505* (0.00000160)		0.00000529* (0.00000154)	0.0000145* (0.00000643)		0.0000136* (0.00000635)
Bureaucratic quality	0.986*** (0.0657)	0.929*** (0.0550)		0.883*** (0.0585)	0.419*** (0.0778)		0.416*** (0.0779)
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	No	No	No	No	Yes	Yes	Yes
Constant	-0.179 (0.203)	0.0316 (0.164)	1.165* (0.569)	0.486* (0.241)	-0.493 (0.256)	0.478 (0.390)	-0.0211 (0.446)
N	43	633	646	633	633	646	633
adj. R ²	0.948	0.920	0.613	0.922	0.240	0.054	0.246

Robust standard errors in parenthesis at 95% confidence intervals: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026b)

Comment: Robustness check changing the IV to females, as a share of public paid employees (wwbi_fspuemp). Model 1 has year 2014 (most observations available), year FE included in model 2-7 (ref. 2000).

Table A 6 Robustnesscheck one-year lag DVI

Dependent variable: Corruption scandals

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6	M7
L.Female senior managers	0.105 (0.246)	-1.023 (0.501)	-0.000 (.)	-1.325 (.)	-1.133* (0.492)	3.099* (1.187)	0.268*** (0.001)
L.Democracy						123.182 (92.924)	31.886*** (0.298)
L. Women in national parliaments (%)						-0.003 (0.991)	0.111*** (0.001)
L.GDP per capita						0.009** (0.002)	0.000*** (0.000)
L.Bureaucratic quality						19.132 (17.003)	2.997*** (0.029)
Country FE	No	No	No	No	Yes	Yes	-
Year FE	No	No	Yes	Yes	Yes	Yes	-
Poisson FE	No	No	No	No	No	No	Yes
Constant	-0.593 (9.032)	48.986** (11.379)	2.000 (.)	50.032 (.)	40.354* (16.276)	-274.038* (96.966)	
Observations	11	12	11	12	23	23	23
Adjusted R ²	-0.101	0.137	.	.	0.803	0.942	

Robust standard errors in parentheses at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Reports One-year lag (L.). Year FE included in all models except 1-2 (ref.2002). Country FE included in FE and Poisson FE in Model 7.

Table A 7 Robustnesscheck five-year lag DVI

Dependent variable: Corruption scandals							
	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6	M7
L5. Female senior manager	-0.143 (0.210)	-3.388*** (0.345)	-14.851 (.)	-3.463 (.)	-3.333*** (0.558)	-5.733* (1.262)	-0.276 (.)
L5. Democracy						134.621 (380.082)	24.246 (.)
L5. Women in national parliaments (%)						-2.522 (1.437)	-0.065 (.)
L5. GDP per capita						0.007 (0.009)	0.001 (.)
L5. Bureaucratic quality						-43.985 (23.953)	-2.738 (.)
Country FE	No	No	No	No	Yes	Yes	-
Year FE	No	No	Yes	Yes	Yes	Yes	-
Poisson FE	No	No	No	No	No	No	Yes
Constant	8.511 (7.786)	96.948*** (6.718)	525.040 (.)	96.782 (.)	121.870*** (18.196)	58.122 (371.663)	
Observations	8	8	8	8	16	16	16
Adjusted R ²	-0.151	0.877	.	.	0.925	0.926	.

Robust standard errors in parentheses at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Reports five-year lag (L5.). Year FE included in all models except 1-2 (ref. 2002). Country FE included in FE and Poisson FE in Model 7.

Table A 8 Robustnesscheck one-year lag DV2

Dependent variable: Female involvement in scandals (%)

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6
L. Female senior manager	0.982	-0.558	1.320	-0.406	-0.714	-4.946**
L.Democracy	(1.344)	(0.489)	(.)	(.)	(0.451)	(1.182)
L.Women in national parliaments (%)						-147.117 (188.645)
L.GDP per capita						-1.606
L.Bureaucratic quality						(1.090)
Country FE	No	No	No	No	Yes	-0.003
Year FE	No	No	Yes	Yes	Yes	(0.003)
Constant	-24.757 (47.964)	34.099* (11.364)	-38.118 (.)	23.681 (.)	30.934 (14.741)	-46.826* (17.370)
Observations	11	12	11	12	23	23
Adjusted R^2	-0.049	0.010	.	.	0.706	0.774

Robust standard errors in parentheses at 95% confidence interval: *** $p \leq 0.001$, ** $p \leq 0.01$, * $p \leq 0.05$.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Reports one-year lag (L.). Year FE included in all models except 1-2 (ref.2002). Country FE included in FE.

Table A 9 Robustnesscheck five-year lag DV2

Dependent variable: Female involvement in scandals (%)

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6
L5. Female senior manager	-2.815**	-0.090	2.673	-1.489	0.599	2.528
	(0.695)	(0.745)	(.)	(.)	(1.074)	(2.389)
L5.Democracy						250.852 (550.976)
L5.Women in national parliaments (%)						0.051 (2.256)
L5.GDP per capita						0.012 (0.007)
L5.Bureaucratic quality						13.626 (31.442)
Country FE	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Constant	112.573** (26.104)	22.216 (13.657)	-84.501 (.)	48.727 (.)	-9.444 (37.314)	-374.269 (505.832)
Observations	8	8	8	8	16	16
Adjusted R ²	0.366	-0.165	.	.	0.502	0.321

Robust standard errors in parentheses at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: Scandal data (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Reports five-year lag (L5). Year FE included in all models except 1-2 (ref.2002). Country FE included in FE.

Table A 10 Robustnesscheck - Logistic regression

Dependent variable: Female involvement in scandals (%)

	M1 (Brazil)	M2 (Spain)	M3 (Brazil)	M4 (Spain)	M5	M6
Female senior managers	-0.990** (0.380)	-0.029 (0.031)	0.474 (1.304)	-0.141 (0.113)	-0.187 ⁺ (0.097)	0.108 (0.391)
Democracy						71.961 (52.404)
Women in parliament						0.876** (0.323)
GDP per capita						-0.001 (0.000)
Bureaucratic quality						8.521 ⁺ (4.972)
Country FE	No	No	No	No	Yes	Yes
Year FE	No	No	Yes	Yes	Yes	Yes
Constant	35.982** (13.793)	0.715 (0.746)	-17.306 (46.487)	3.468 (2.755)	6.440* (3.172)	-66.825 (53.773)
<i>N</i>	41	308	28	308	349	349
Pseudo R ²	0.15	0.002	0.095	0.027	0.031	0.043

Robust standard errors in parenthesis at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: Scandal data 2002-2014 (Riberio et al., 2018), QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Development Indicators (WDI 2026). Comment: Logistic regression with margins. Logit models report log-odds. Year FE included in models 3-6 (ref.2002), Country FE in model 5-6.

Table A 11 Cross-sectional models on the effect of female managers on corruption

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M4	M5	M6	M7
Female managers in public sector (%)	- 1.605 (0.808)	5.196* (1.287)	1.106* (2.08)	-0.579 (0.361)	5.812* ** (1.410)	0.147** (0.0448)	-0.519 (0.412)	1.795** * (3.92)
Electoral democracy		7.455* ** (1.154)	0.887 (1.52)	-0.313 (0.352)	7.097* ** (1.193)	0.567* (0.248)	0.0950 (0.352)	1.814** (2.92)
Female managers*Electoral democracy		-9.533 *** (2.627)	-1.110 (-0.86)		-10.07 *** (2.627)			- 3.425** * (-3.76)
Women in parliament			-0.00716 (-1.74)	-0.00664 (0.00505)		-0.00122 (0.00169)	-0.00649 (0.00530)	-0.00719 (-1.48)
GDP per capita			0.00000830 *** (4.90)	0.00000725 *** (0.00000248)		00000607 (0.00000388)	0.000005 41 (0.00000248)	0.00000 519 (1.62)
Bureaucratic quality			0.882*** (9.75)	1.011*** (0.0855)		0.389*** (0.0569)	0.960*** (0.133)	0.904** * (7.52)
Constant	0.781 (0.406)	- 4.562* * (0.610)	-0.902* (-2.43)	0.232 (0.309)	-4.311 (0.758)	-0.766 (0.525)	0.046 (0.392)	-1.135** (-3.30)
N	46	46	44	32	34	32	32	32
adj. R ²	0.041	0.623	0.95	0.936	0.047	0.226	0.90	0.92

Robust standard errors in parenthesis at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) and World Bank Bureaucracy Indicators (World Bank 2024). Comment: Bivariate regression model 1-4, bivariate regression including time-fixed effects Model 3-4. Model 1 has year 2014 (most observations available). Model 4-5 include one-year lag, Model 5-7 include five-year lag.

Table A 12 First-difference models on the effect of female managers on corruption

Dependent variable: Control of Corruption

	M1	M2	M3	M4	M5
Female managers in public sector (%)	0.105 (0.0524)	0.135* (0.0512)	0.147** (0.0448)	0.346 (0.220)	0.319 (0.220)
Electoral democracy			0.567* (0.248)	0.462** (0.158)	0.646* (0.259)
Female managers*Electoral democracy				-0.302 (0.283)	-0.230 (0.280)
Women in parliament			-0.00122 (0.00169)		-0.001 (0.002)
GDP per capita			0.0000607 (0.00000388)		0.000 (0.000)
Bureaucratic quality			0.389*** (0.0569)		0.387*** (0.057)
Year FE	No	Yes	Yes	Yes	Yes
Country FE	No	No	No	No	No
Constant					
N	457	457	447	475	447
adj. R ²	0.005	0.047	0.226	0.053	0.225

Robust standard errors in parenthesis at 95% confidence interval: ***p≤0.001, **p≤0.01, *p≤0.05.

Source: QoG Standard TS Dataset Version Jan26 (Teorell et al., 2026) & World Bank Bureaucracy Indicators (World Bank 2024).Comment: First-Difference models. Bivariate regression model 1-2. Model 2-5 include year FE (ref.2000).

A13. Potential Outcomes Framework

The average treatment effect (ATE) is a causal estimate based on a contrast of potential outcomes (Keele 2015: 316). In this thesis, the causal effect and potential outcomes framework draws upon a non-binary treatment between women in top bureaucratic positions and corruption.

$Y(d)$ equals the level of corruption when the share of women in top bureaucratic positions is d , and $Y(d+1)$ equals the levels of control of corruption when the share of women in top bureaucratic positions increase with one-unit.

The estimated difference is $Y(d+1) - Y(d)$, and the causal effect $E[Y_{(d+1)} - Y_{(d)}]$ suggest the average effect of a one-unit increase in the share of women in top bureaucratic positions on the average change in levels of control of corruption.

The counterfactual potential outcomes framework $E[Y_{(d)}|D = d+1] = E[Y_{(d)}|D = d]$ suggest that in the absence of a one-unit increase in the share of women in top bureaucratic positions, the average change in control of corruption in units (e.g countries in this case) that experienced a one-unit increase in the share of women would have been the same as those observations that did not experience a one-unit increase.

The analysis relies on linear regression with two-way fixed effects (year and country FE). Hence, the potential outcomes notation follows $Y_{it}(d)$, the level of corruption in country i at time t , and d_1 if the share of women increases by one-unit.

The average causal effect with fixed effects suggests $(d_1, d_0) = E[Y_{it}(d_1)|\alpha_i, \gamma_t] - E[Y_{it}(d_0)|\alpha_i, \gamma_t]$, α_i = the country fixed effects and γ_t = year fixed effects, equals increasing the share of women in top bureaucratic positions by one-unit on the average change in levels of control of corruption, controlling for time-invariant and time-variant factors.

A14. Distribution of IV per country per year ⁶

code	year	female_ma_merger	code	year	female_ma_merger	code	year	female_ma_merger	code	year	female_ma_merger	code	year	female_ma_merger	code	year	female_ma_merger
8	2002	65759033 Matched (3)	40	2005	31153136 Matched (3)	76	2002	34945703 Matched (3)	144	2002	2817238 Matched (3)	188	2007	5251742 Matched (3)	218	2014	04820395 Matched (3)
8	2005	67145208 Matched (3)	40	2006	32360398 Matched (3)	76	2003	34366268 Matched (3)	144	2004	45591518 Matched (3)	188	2008	60763252 Matched (3)	218	2015	06718202 Matched (3)
24	2000	51152992 Matched (3)	40	2007	39019415 Matched (3)	76	2004	36947152 Matched (3)	144	2006	20551792 Matched (3)	188	2009	5291245 Matched (3)	218	2016	06959396 Matched (3)
24	2008	51614116 Matched (3)	40	2008	18977897 Matched (3)	76	2005	3535925 Matched (3)	144	2007	12033612 Matched (3)	188	2010	53016311 Matched (3)	218	2017	33561962 Matched (3)
24	2014	27815807 Matched (3)	40	2009	3770262 Matched (3)	76	2006	36263388 Matched (3)	144	2008	15186718 Matched (3)	188	2012	41973001 Matched (3)	222	2000	35802704 Matched (3)
31	2011	58059788 Matched (3)	40	2010	260066 Matched (3)	76	2007	34456074 Matched (3)	144	2009	23914051 Matched (3)	188	2015	543085 Matched (3)	222	2001	50538408 Matched (3)
32	2003	4831585 Matched (3)	40	2011	33328232 Matched (3)	76	2008	29455835 Matched (3)	144	2011	17695239 Matched (3)	188	2016	37267518 Matched (3)	222	2003	40348792 Matched (3)
32	2004	42677232 Matched (3)	40	2012	35430473 Matched (3)	76	2009	26369966 Matched (3)	144	2012	1871755 Matched (3)	188	2021	40792588 Matched (3)	222	2004	31214752 Matched (3)
32	2005	54667455 Matched (3)	40	2013	3244859 Matched (3)	76	2011	32255771 Matched (3)	144	2013	280213 Matched (3)	191	2010	3978693 Matched (3)	222	2006	17962362 Matched (3)
32	2006	51112211 Matched (3)	40	2014	32303378 Matched (3)	76	2012	24874978 Matched (3)	144	2014	33902666 Matched (3)	191	2011	24133705 Matched (3)	222	2007	31534842 Matched (3)
32	2007	47087207 Matched (3)	40	2015	32099535 Matched (3)	76	2014	23202103 Matched (3)	144	2015	23813485 Matched (3)	191	2012	28788567 Matched (3)	222	2008	25187933 Matched (3)
32	2008	45089576 Matched (3)	40	2016	29118663 Matched (3)	76	2015	23329966 Matched (3)	144	2016	3412146 Matched (3)	191	2013	20293447 Matched (3)	222	2009	30632234 Matched (3)
32	2009	4108015 Matched (3)	40	2017	33806758 Matched (3)	76	2016	1831207 Matched (3)	148	2017	63858576 Matched (3)	191	2014	36259004 Matched (3)	222	2012	43583836 Matched (3)
32	2010	45268135 Matched (3)	40	2018	25022303 Matched (3)	76	2017	19867906 Matched (3)	152	2000	23625378 Matched (3)	191	2015	35703403 Matched (3)	222	2014	25512967 Matched (3)
32	2011	45019996 Matched (3)	50	2010	42728277 Matched (3)	76	2021	20696836 Matched (3)	152	2003	21615627 Matched (3)	191	2016	33962381 Matched (3)	222	2015	36895517 Matched (3)
32	2012	56903857 Matched (3)	50	2013	57498709 Matched (3)	100	2007	28766814 Matched (3)	152	2006	26890993 Matched (3)	191	2017	31376201 Matched (3)	222	2016	45675012 Matched (3)
32	2013	47427261 Matched (3)	50	2015	29589155 Matched (3)	100	2008	19003598 Matched (3)	152	2009	26524885 Matched (3)	191	2018	37789166 Matched (3)	222	2017	44438892 Matched (3)
32	2014	43414468 Matched (3)	56	2007	42544684 Matched (3)	100	2014	3230882 Matched (3)	170	2002	60165176 Matched (3)	196	2010	62229377 Matched (3)	231	2006	87210474 Matched (3)
32	2015	42456797 Matched (3)	51	2021	34467933 Matched (3)	100	2010	2178845 Matched (3)	152	2013	1096182 Matched (3)	196	2006	61614513 Matched (3)	231	2009	88033975 Matched (3)
32	2016	39411816 Matched (3)	56	2004	3642413 Matched (3)	100	2011	26359925 Matched (3)	152	2015	13301282 Matched (3)	196	2007	46919119 Matched (3)	231	2003	88420892 Matched (3)
32	2017	46973988 Matched (3)	56	2005	48513937 Matched (3)	100	2012	35494846 Matched (3)	152	2017	24735578 Matched (3)	196	2008	6352492 Matched (3)	231	2004	88524717 Matched (3)
32	2018	46405077 Matched (3)	56	2006	4695406 Matched (3)	100	2013	39762768 Matched (3)	170	2001	57419658 Matched (3)	196	2009	64194977 Matched (3)	231	2005	91153352 Matched (3)
36	2001	3353729 Matched (3)	56	2007	42544684 Matched (3)	100	2014	3230882 Matched (3)	170	2002	60165176 Matched (3)	196	2010	62229377 Matched (3)	231	2006	87210474 Matched (3)
36	2002	3156752 Matched (3)	56	2008	39304689 Matched (3)	100	2015	30424166 Matched (3)	170	2004	63042966 Matched (3)	196	2011	50367856 Matched (3)	231	2009	81121916 Matched (3)
36	2003	34819072 Matched (3)	56	2009	45482805 Matched (3)	100	2016	31546637 Matched (3)	170	2005	60792696 Matched (3)	196	2012	51388301 Matched (3)	231	2010	7586599 Matched (3)
36	2004	34819072 Matched (3)	56	2010	4130429 Matched (3)	100	2017	3678454 Matched (3)	170	2006	56332155 Matched (3)	196	2013	60620636 Matched (3)	231	2011	90219027 Matched (3)
36	2005	34819072 Matched (3)	56	2011	35894726 Matched (3)	100	2018	29984856 Matched (3)	170	2009	57306224 Matched (3)	196	2014	51761976 Matched (3)	231	2012	77598614 Matched (3)
36	2006	37673845 Matched (3)	56	2012	2684192 Matched (3)	100	2019	2684192 Matched (3)	170	2010	52048012 Matched (3)	196	2015	52048012 Matched (3)	231	2013	82658012 Matched (3)
36	2007	37673845 Matched (3)	56	2013	23800102 Matched (3)	116	2003	5257206 Matched (3)	170	2011	53352676 Matched (3)	196	2017	27675235 Matched (3)	231	2015	7560429 Matched (3)
36	2008	37673845 Matched (3)	56	2014	31304878 Matched (3)	116	2006	7545585 Matched (3)	170	2012	52399926 Matched (3)	208	2016	2956706 Matched (3)	231	2016	79414564 Matched (3)
36	2009	37673845 Matched (3)	56	2015	34542897 Matched (3)	116	2008	70181662 Matched (3)	170	2014	4739845 Matched (3)	208	2017	34735004 Matched (3)	231	2021	83247236 Matched (3)
36	2010	40390351 Matched (3)	56	2016	37330337 Matched (3)	116	2009	75790472 Matched (3)	170	2015	4739845 Matched (3)	208	2018	4739845 Matched (3)	231	2004	39802636 Matched (3)
36	2011	40390351 Matched (3)	56	2017	34524 Matched (3)	116	2012	6471712 Matched (3)	188	2001	54880542 Matched (3)	218	2000	23264921 Matched (3)	233	2005	42333643 Matched (3)
36	2012	2855963 Matched (3)	56	2018	34118792 Matched (3)	120	2014	26190722 Matched (3)	188	2002	48448688 Matched (3)	218	2004	28110548 Matched (3)	233	2006	42875826 Matched (3)
36	2013	24398267 Matched (3)	64	2003	4611986 Matched (3)	120	2022	30282319 Matched (3)	188	2003	50352997 Matched (3)	218	2005	19099184 Matched (3)	233	2007	31677166 Matched (3)
36	2014	28569494 Matched (3)	64	2007	35016236 Matched (3)	132	2000	38749998 Matched (3)	188	2004	50252932 Matched (3)	218	2006	27320883 Matched (3)	233	2008	24000212 Matched (3)
36	2015	28569494 Matched (3)	64	2009	43936625 Matched (3)	132	2001	43936625 Matched (3)	188	2005	43936625 Matched (3)	218	2007	43936625 Matched (3)	233	2009	43936625 Matched (3)
36	2016	24123615 Matched (3)	72	2015	59289856 Matched (3)	144	2000	40311202 Matched (3)	188	2006	45167938 Matched (3)	218	2013	05942895 Matched (3)	233	2010	24200156 Matched (3)
246	2008	24357225 Matched (3)	262	2015	81621395 Matched (3)	320	2004	33776104 Matched (3)	348	2004	49624965 Matched (3)	352	2012	57304660 Matched (3)	380	2010	23394455 Matched (3)
246	2012	32979542 Matched (3)	268	2008	66166604 Matched (3)	320	2006	09686502 Matched (3)	348	2005	41831445 Matched (3)	352	2013	61172712 Matched (3)	380	2011	36524636 Matched (3)
246	2013	3514941 Matched (3)	268	2009	69914885 Matched (3)	320	2011	21562426 Matched (3)	348	2006	40599906 Matched (3)	356	2011	52134907 Matched (3)	380	2012	45182696 Matched (3)
246	2014	3514941 Matched (3)	268	2010	63382119 Matched (3)	320	2012	2684273 Matched (3)	348	2007	41831445 Matched (3)	356	2012	45182696 Matched (3)	380	2013	45182696 Matched (3)
246	2015	43727453 Matched (3)	268	2012	51043435 Matched (3)	340	2001	36148426 Matched (3)	348	2008	3644422 Matched (3)	356	2019	38246223 Matched (3)	380	2014	53738575 Matched (3)
246	2016	52056195 Matched (3)	268	2013	43254033 Matched (3)	340	2002	2269052 Matched (3)	348	2009	4484781 Matched (3)	360	2018	58688811 Matched (3)	380	2015	40556188 Matched (3)
246	2017	43880858 Matched (3)	288	2005	51081747 Matched (3)	340	2003	28223103 Matched (3)	348	2010	45586723 Matched (3)	372	2004	15883783 Matched (3)	380	2016	45620954 Matched (3)
246	2018	36952266 Matched (3)	288	2012	5814752 Matched (3)	340	2014	3456538 Matched (3)	348	2011	46158652 Matched (3)	372	2005	13169592 Matched (3)	380	2017	31669837 Matched (3)
250	2004	1494904 Matched (3)	288	2016	63370139 Matched (3)	340	2016	29670712 Matched (3)	348	2012	5434237 Matched (3)	372	2006	19919635 Matched (3)	380	2018	3313274 Matched (3)
250	2005	16507063 Matched (3)	300	2007	47016633 Matched (3)	340	2006	34358242 Matched (3)	348	2013	4922170 Matched (3)	372	2007	16843355 Matched (3)	388	2001	3393334 Matched (3)
250	2006	18355568 Matched (3)	300	2008	56663263 Matched (3)	340	2007	3312454 Matched (3)	348	2014	44504565 Matched (3)	372	2008	15161863 Matched (3)	388	2002	15161863 Matched (3)
250	2007	17761257 Matched (3)	300	2009	5839099 Matched (3)	340	2008	3558567 Matched (3)	348	2015	55840655 Matched (3)	372	2009	15959531 Matched (3)	400	2002	5258832 Matched (3)
250	2008	18839176 Matched (3)	300	2010	2010183 Matched (3)	340	2009	3816829 Matched (3)	348	2016	57612613 Matched (3)	372	2010	1467517 Matched (3)	400	2003	70545816 Matched (3)
250	2009	18929263 Matched (3)	300	2011	388273 Matched (3)	340	2010	27913828 Matched (3)	348	2017	49914703 Matched (3)	372	2011	2020634 Matched (3)	400	2011	69697645 Matched (3)
250	2010	18747897 Matched (3)	300	2012	46861714 Matched (3)	340	2011	3171316 Matched (3)	348	2018	4300811 Matched (3)	372	2012	20642478 Matched (3)	400	2012	6934014 Matched (3)
250	2011	29473126 Matched (3)	300	2013	467379 Matched (3)	340	2012	2884157 Matched (3)	352	2004	49570826 Matched (3)	372	2013	26640551 Matched (3)	400	2013	7319078 Matched (3)
250	2012	30534792 Matched (3)	300	2014	51137485 Matched (3)	340	2013	28055641 Matched (3)	352	2005	49285927 Matched (3)	372	2014	29322333 Matched (3)	400	2014	70931697 Matched (3)
250	2013	36494904 Matched (3)	300	2015	45810731 Matched (3)	340	2014	3816829 Matched (3)	352	2006	4998465 Matched (3)	372	2015	49763078 Matched (3)	400	2015	87137206 Matched (3)
250	2014	31791215 Matched (3)	300	2017	62819225 Matched (3)	340	2015	32776445 Matched (3)	352	2007	49217203 Matched (3)	372	2016	35805762 Matched (3)	404	2005	47802675 Matched (3)
250	2015	35708892 Matched (3)	300	2018	66978228 Matched (3)	340	2016	33659422 Matched (3)	352	2008	39526638 Matched (3)	372	2017	3883923 Matched (3)	426	2010	59173996 Matched (3)
250	2016	3198013 Matched (3)	320	2000	15397517 Matched (3)	340	2017	15367618 Matched (3)	352	2009	51500148 Matched (3)	380	2007	35013056 Matched (3)	426	2018	87664552 Matched (3)

